CHAPTER 1

INTRODUCTION

This chapter introduces the concept of the Site Material Management System and provides an overview of the sponsored capstone project. It explains the basic idea of construction material management operations, including how materials are purchased, stored, issued, and tracked at the construction site. The chapter also describes the key activities involved in the system and the overall material management process flow.

The purpose of this chapter is to build a clear understanding of how construction site material management works and how a digital system can improve its efficiency. It highlights the importance of proper management of inventory, material usage, supplier details, and transaction records.

Site's Material Management System:-

The construction site selected for this project is involved in building and infrastructure development activities. Construction work requires a continuous supply of materials such as cement, steel, sand, bricks, and other essential resources. Managing these materials manually becomes difficult as the project size increases.

At most construction sites, material records are maintained using registers or basic spreadsheets. This leads to issues like incorrect stock tracking, material wastage, and delays in project execution. To overcome these challenges, a digital system is required.

The Site Material Management System is developed to provide a structured and efficient way of handling materials, ensuring proper tracking and better utilization of resources

Material Management Work :-

Material management is the process of planning, organizing, and controlling the flow of materials from purchase to usage at the construction site. It ensures that the right materials are available at the right time and in the right quantity.

In real-life construction projects, proper material management is very important because it directly affects project cost, quality, and completion time. A well-managed system helps in reducing wastage, avoiding shortages, and improving overall productivity.

Material management involves several key activities and processes to ensure smooth operations and effective control.

These include:

Material and Category Management: All materials must be properly listed with category-wise details so that they can be easily identified and managed. Proper classification helps in better stock organization and quick access.

Purchase Management: Every material purchase must record supplier details, material quantity, cost, and purchase date. Proper purchase management ensures transparency and accurate tracking of incoming materials.

Stock and Inventory Tracking: Real-time tracking of material availability is essential. It helps staff know which materials are available, which are in use, and which need to be reordered.

Material Issue and Usage Tracking: Materials issued for construction work must be recorded properly. This helps in tracking how much material is used and prevents misuse or wastage.

Inspection and Storage Handling: Materials are inspected at the time of receiving to ensure quality. Proper storage is also important to avoid damage and maintain material condition.

Reporting and Monitoring: Regular reports help in analyzing material usage, stock levels, and future requirements. This improves decision-making and planning.

CHAPTER 2

LITERATURE SURVEY

This chapter presents the study of existing material management systems used at construction sites. It focuses on understanding how current systems operate, including manual record-keeping methods and basic digital tools used for managing materials. The chapter highlights the limitations and problems faced in these systems, which affect efficiency, accuracy, and overall project performance.

The purpose of this study is to analyze real-world issues such as improper stock tracking, delay in material availability, poor record management, and difficulty in monitoring material usage. This analysis helps in identifying the need for an improved Site Material Management System.

Requirement Gathering:

Questions Asked in Site Supervisor:

Q1. Do you face issues like incorrect stock records due to manual updates?
Answer: Yes, sometimes stock records are not updated properly, which leads to incorrect material availability.

Q2. Do you experience delays in checking material availability?
Answer: Yes, it takes time to check stock manually, especially when records are maintained in registers

Q3. Is there any confusion in tracking material quantity or usage?
Answer: Yes, sometimes it is difficult to track how much material is used or remaining.

Q4. Do you face problems in managing material purchases and supplier records?
Answer: Yes, managing supplier details and purchase records manually is difficult and time-consuming.

Q5. Is it difficult to track previous material records and history?
Answer: Yes, searching old records in registers takes a lot of time.

Q6. Do you have a proper system for monitoring material usage and wastage?
Answer: No, there is no proper system; everything is managed manually.

Q7. Do you find it difficult to check real-time stock availability?
Answer: Yes, real-time tracking is not possible in manual systems.

Q8. Is managing material details manually time-consuming?
Answer: Yes, maintaining records manually requires a lot of time and effort.

Q9. Do you face issues while issuing materials for site work?
Answer: Yes, sometimes material issue records are not properly maintained, which creates confusion later.

This study clearly shows that existing systems are mostly manual and inefficient. The problems identified highlight the need for a computerized Site Material Management System that can improve accuracy, reduce workload, and provide real-time tracking of materials.

2.1 Study of Existing system

This section presents the study of existing material management systems used at construction sites. These systems include manual processes as well as partially digital tools such as spreadsheets. The study focuses on understanding their working methods, features, and limitations.

2.2.1 Manual Process System

Currently, most construction sites manage materials using a manual system. In this system, material details such as stock, purchase records, supplier information, and material usage are maintained in paper registers or basic Excel sheets. Communication between site supervisors and store managers is usually done through phone calls or verbal instructions.

This system is highly dependent on manual work, which increases the chances of errors and reduces overall efficiency. Managing large quantities of materials becomes difficult, especially in bigger construction projects.

Key Issues in Manual System:

- High chances of human error in stock and record keeping
- Difficulty in searching old material records

- No real-time stock availability tracking
- Risk of data loss due to damage or misplacement of registers
- Time-consuming process

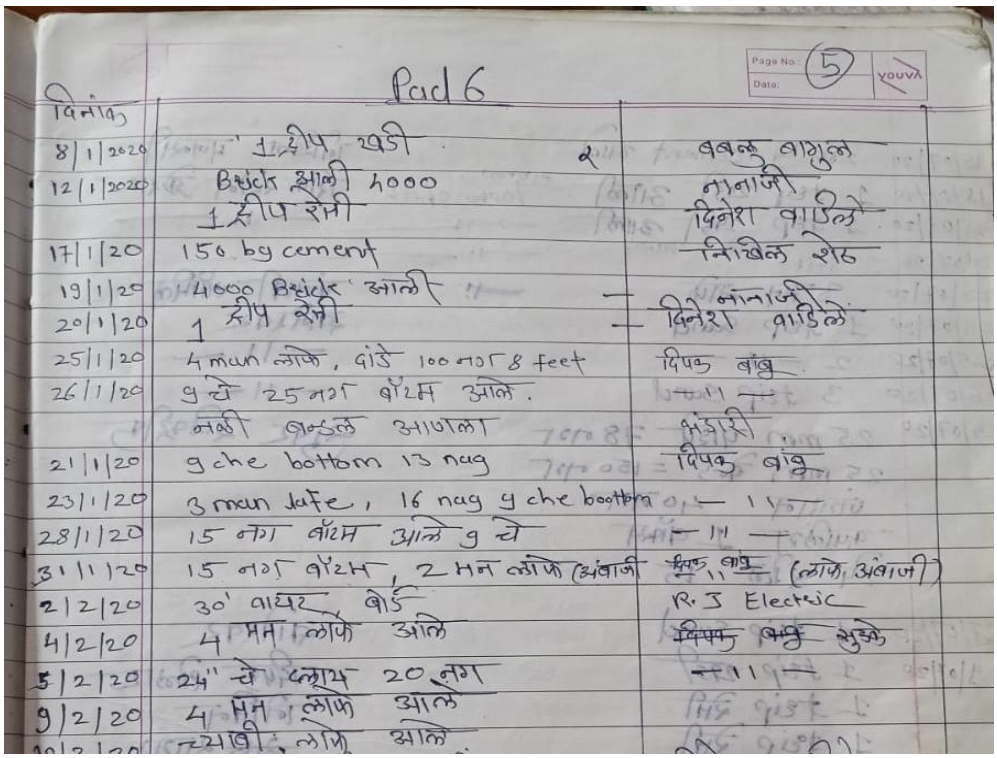

Fig.2.1 Manual Material Record

As shown in Fig. 2.1, booking records are maintained manually using registers, which makes data management difficult and time-consuming.

2.2.2 Issues for Double booking in Manual System

Incorrect stock management is one of the most common and serious problems in manual material management systems. It occurs when materials are issued without proper record updates or when the same stock is considered available multiple times due to lack of real-time tracking. This usually happens because records are not updated immediately or are maintained incorrectly.

Effects of Stock Mismanagement:

- Shortage of materials during construction work
- Delay in project execution
- Confusion among site staff

- Increased material wastage or misuse

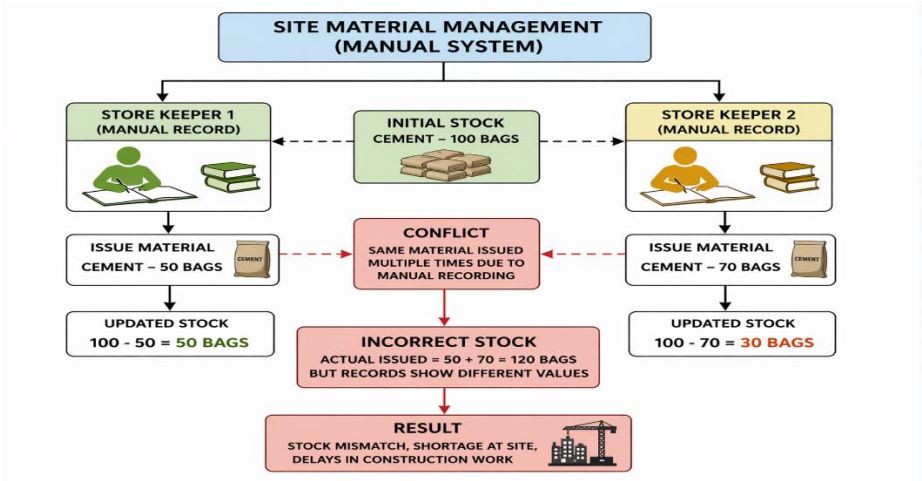

Fig.2.2 Incorrect Material Issue / Stock Conflict

As shown in Fig. 2.2, the same material is issued multiple times by different storekeepers without proper updating of records in the manual system. Due to the absence of real-time stock tracking, both users assume that sufficient stock is available and issue materials independently. This results in incorrect stock calculations and mismatch between actual and recorded quantities.

This problem leads to confusion in material management, shortage of materials at the construction site, and delays in ongoing work. It reduces system reliability and negatively affects overall project efficiency.

2.2.3 Material Categories Structure

Materials in the system are divided into categories for easy management and identification.

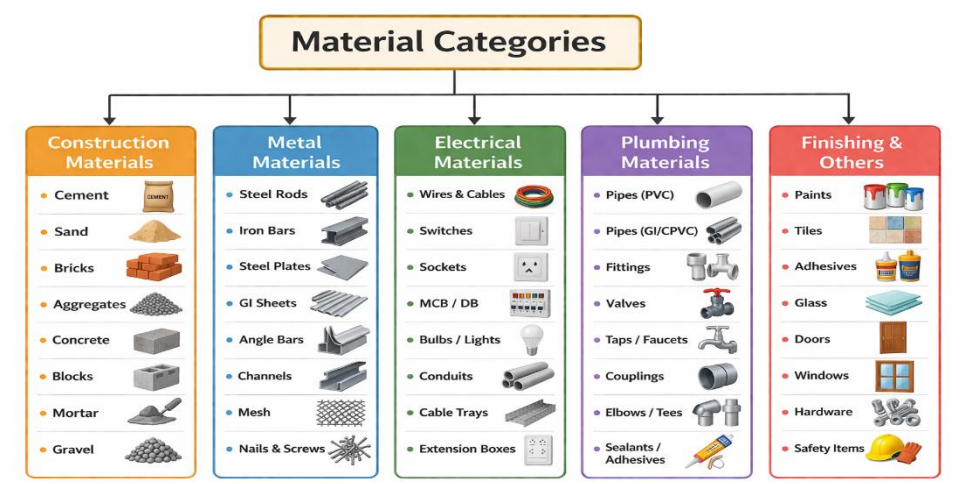

Fig.2.3 Material Categories Structure

Materials in the existing system are organized into different categories such as construction materials, metal materials, electrical materials, plumbing materials, and finishing materials. Each category is further divided into subcategories based on type and usage. This structure helps users easily identify, manage, and track materials at the construction site.

Based on the study of existing platforms, the following table summarizes the key limitations and problems identified across major functional areas.

Limitations of Existing Material Systems:-

- Incorrect stock levels due to manual updates and lack of real-time tracking
- Delay and confusion in issuing materials because of manual record-keeping
- Lack of clarity in actual material usage and remaining stock
- Difficulty in managing purchase records due to absence of a structured system
- Poor tracking of material history and lack of proper documentation
- No proper inspection records, leading to the use of low-quality materials
- Absence of an effective system to monitor material usage and prevent wastage

2.2 Problem Identification

After studying the existing systems, the following problems were clearly identified for both users and admin/site manager stakeholders:

The system involves key stakeholders such as users (staff) and admin/site managers, each facing different challenges in the existing manual system. Users experience difficulty in checking real-time material availability due to reliance on manual records. This often leads to confusion regarding stock quantities and actual material usage. Additionally, delays occur while issuing materials because there is no proper automated system in place.

On the other hand, Admins or Site Managers face issues in managing overall material operations. Manual handling of records increases the chances of errors and delays in daily activities. They also find it difficult to manage stock levels, purchase records, and related details efficiently. The absence of a proper digital system results in poor record maintenance, increasing the risk of material mismanagement and lack of control over site operations.

2.3 Needs of system

Based on the problems identified above, the following table outlines the functional requirements and needs of the proposed Site Material Management System:

The proposed system is designed to fulfill the needs of both users and admin/site managers involved in material handling operations. For users, the system provides easy access to material details and stock information, enabling them to quickly understand availability. It also offers real-time stock tracking, which helps in avoiding shortages or over-issuing of materials.

For Admins and Site Managers, the system provides a user-friendly dashboard to efficiently manage materials, stock, and overall operations. It also maintains digital records of all material transactions and history, reducing dependency on manual processes. Furthermore, the system includes proper monitoring features to control material usage and minimize wastage, ensuring better resource management and operational efficiency.

2.4 Problem Definition

Managing construction materials through a system is challenging because it must accurately track material stock, purchases, usage, and records. Many existing systems still face problems such as:

- Incorrect stock updates causing material shortage or over-issue
- Delays in issuing materials due to manual record handling
- Manual or unclear record tracking for material transactions
- Confusion in tracking material quantity and usage
- Difficulty in managing multiple materials and site requirements simultaneously

Therefore, we propose a Site Material Management System that

- Structured Material Management: Maintains organized and detailed records of all materials with proper categorization
- Real-Time Stock Tracking: Updates stock automatically to ensure accurate availability of materials
- Material Issue Tracking: Tracks material usage and issue records properly
- Digital Records: Stores all material transactions and history digitally for better management
- Efficiency: Reduces manual effort and improves overall efficiency for site managers and users

This chapter analyzed the existing material management systems and identified their major limitations, including manual errors, lack of real-time tracking, and inefficient record management. It also highlighted the key problems faced by both users and site administrators. Based on this study, the need for a structured and digital material management system was clearly established. The next chapter describes the scope of the proposed system and defines its functional boundaries and user roles.

CHAPTER 3

SCOPE OF THE PROJECT

This chapter defines the scope of the Site Material Management System and outlines the functionalities included in the project. It explains how the system replaces manual material handling operations with a structured digital approach. The chapter also describes the roles and responsibilities of Admin, Site Manager/Staff, and User in the system.

The objective of this chapter is to clearly define what the system will do and what is excluded from the project.

- Improving day-to-day material management operations
- Providing real-time stock availability
- Tracking material usage and stock updates
- Generating useful reports for monitoring and decision-making

3.1 Functional Scope

The functional scope of the proposed system covers the complete material management lifecycle, starting from material entry to usage tracking and final reporting. It includes all major operations required to manage construction site materials efficiently, ensuring accuracy, transparency, and better resource utilization.

3.1.1 Material and Category Management

The system enables the administrator to manage materials and categories in a structured manner.

- Add, update, and delete material categories
- Maintain material details such as name, type, quantity, unit, and cost
- Organize materials category-wise for easy access
- Track real-time availability of materials

3.1.2 Stock Management

- Update stock when new materials are purchased
- Deduct stock when materials are issued
- Maintain real-time stock availability

- Prevent over-issue of materials

3.1.3 Supplier Management

This module manages supplier-related information.

- Add and maintain supplier details
- Record materials supplied by each supplier
- Maintain contact details for communication
- Track supplier history

3.1.4 Material Issue Management

This module handles issuing of materials for site work.

- Record material issue details (quantity, date, user)
- Track material usage at the site
- Prevent duplicate or excess issue
- Maintain issue history

3.1.5 Purchase Management

This module handles issuing of materials for site work.

- Record material issue details (quantity, date, user)
- Track material usage at the site
- Prevent duplicate or excess issue
- Maintain issue history

3.1.6 Reporting and Dashboard

The system provides reports for monitoring and decision-making.

- Generate stock reports
- Track material usage reports
- View purchase and supplier reports
- Analyze stock availability and shortages

3.2 In-Scope and Out-of-Scope

The proposed Site Material Management System is designed to handle essential material-related operations required at a construction site. The system includes functionalities such as material entry, category management, stock tracking, and real-time availability monitoring. It allows users to issue materials, record usage, and maintain proper transaction records, which helps in improving accuracy and reducing manual errors. The system also

supports purchase record management, enabling better tracking of material costs and stock

3.3 User Requirements

The system is designed to fulfil the requirements of different types of users, including Admin and Staff (Site Supervisor/Store Manager). Each user has specific roles and responsibilities within the Site Material Management System.

3.1.1 Admin

The Admin is responsible for overall control, configuration, and monitoring of the system.

- **User Management:** Add, update, or remove users and assign roles
- **Material Management:** Manage material categories and material details
- **Stock Supervision:** Monitor stock levels and material availability
- **Purchase Oversight:** Verify purchase records and supplier details
- **Reporting:** Access complete reports across all modules
- **System Control:** Ensure data accuracy and smooth system operation

3.1.2 Staff (Site Supervisor / Store Manager)

Staff members are responsible for handling day-to-day material operations.

- **Material Entry:** Add and update material details
- **Stock Handling:** Check and update material availability
- **Material Issue:** Issue materials for site work and record usage
- **Purchase Entry:** Record material purchases and supplier details
- **Tracking:** Monitor material usage and stock status
- **Basic Reports:** View operational reports as required

3.2 Non-Functional Scope

The system also includes non-functional requirements to ensure quality, performance, and usability.

- **Performance:** Fast response time for material entry, issue, and tracking
- **Reliability:** Consistent and accurate data across all modules
- **Security:** Role-based access control and secure login system
- **Usability:** Simple and user-friendly interface for all users

- **Scalability:** Ability to handle growth in materials

3.3 Expected Outcomes of Project Scope

By implementing the proposed system, the following outcomes are expected:

- Improved control over material stock and availability
- Reduction in manual errors in stock and issue record
- Faster and more accurate material tracking
- Better utilization of construction resources
- Enhanced efficiency in site operations
- Accurate and reliable reports for decision-making

This chapter explained the functional and non-functional scope of the system along with user requirements and expected outcomes. It defined the responsibilities of different users and ensured that the system remains efficient, accurate, and aligned with construction site requirements. The next chapter discusses the methodology used for designing and developing the system.

CHAPTER 4

METHODOLOGY

This chapter explains the methodology used to design and develop the Site Material Management System. It describes the structured approach followed during development, including requirement analysis, system design, implementation, and testing.

The project follows the Waterfall Model of SDLC to ensure systematic and controlled development. The purpose of this chapter is to provide a clear understanding of how the system is planned and implemented step by step.

To maintain discipline in development, each phase is completed and verified before moving to the next phase.

4.1 SDLC Approach Followed Waterfall Model:

The Waterfall Model is selected because the system requirements are clearly defined and can be implemented in a sequential manner. Each phase produces proper outputs, ensuring better control and accuracy.

Phases followed:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

The following diagram shows the sequential phases of the Waterfall Model adopted for this project:

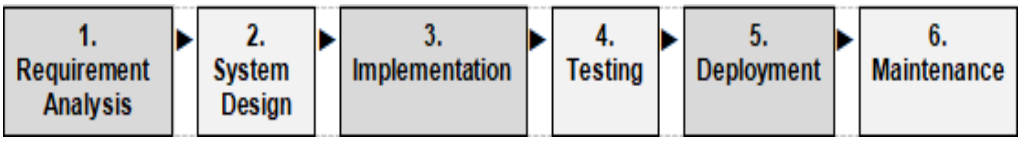

Figure 4.1: SDLC Waterfall Model — Sequential Phase Flow

4.2 Requirement Analysis

Requirement analysis was carried out through observation and discussions with site staff to understand the actual needs of material management at construction sites.

- Discussion with site supervisors and store managers
- Observation of manual stock handling and material issue process
- Study of common errors in manual record keeping
- Identification of required modules based on user roles

Key Problems Identified in Existing / Manual Workflow

- Material stock is not updated in real time
- Difficulty in tracking issued and remaining materials
- Errors in manual record keeping
- Poor tracking of material usage and wastage
- Delayed and inaccurate reporting
- Lack of proper monitoring of material flow

4.2.1 Admin Requirements

The Admin manages and controls the overall system operations. Material stock is not updated in real time

- Add and manage users with role assignments
- Maintain material categories and material details
- Monitor stock levels and material usage
- Access reports and dashboard for decision-making
- Ensure system security and data accuracy

4.2.2 Staff Requirements (Site Supervisor / Store Manager)

Staff members handle daily material operations at the site

- Add and update material records
- Check material availability before issuing
- Record material issue and usage
- Update stock after material consumption

- Maintain purchase and supplier records
- Track material movement and usage history

4.3 Staff Requirements (Site Supervisor / Store Manager)

Staff interact with the system to manage materials, track usage, and maintain stock records. Their requirements focus on efficiency, accuracy, and real-time updates.

- Add and update material details
- Check material availability before issuing
- Issue materials for site work
- Track material usage and stock updates
- View material status and availability

4.4 System Design Methodology

The system design phase focuses on creating a structured plan for implementing the Site Material Management System. This phase includes designing the system architecture, database structure, and user interface based on the requirements gathered from Admin and Staff.

Additionally, the system design helps improve overall efficiency by reducing manual errors and ensuring accurate data handling. It also supports real-time updates of material availability and stock levels, which improves site operations.

The system is designed to be scalable so that new features can be added in the future. Proper validation and security mechanisms are included to protect data and ensure reliable performance

Core Functionalities of the System:

- Material Management
- Stock Management
- Material Issue & Usage Tracking
- Purchase Management

- User Management

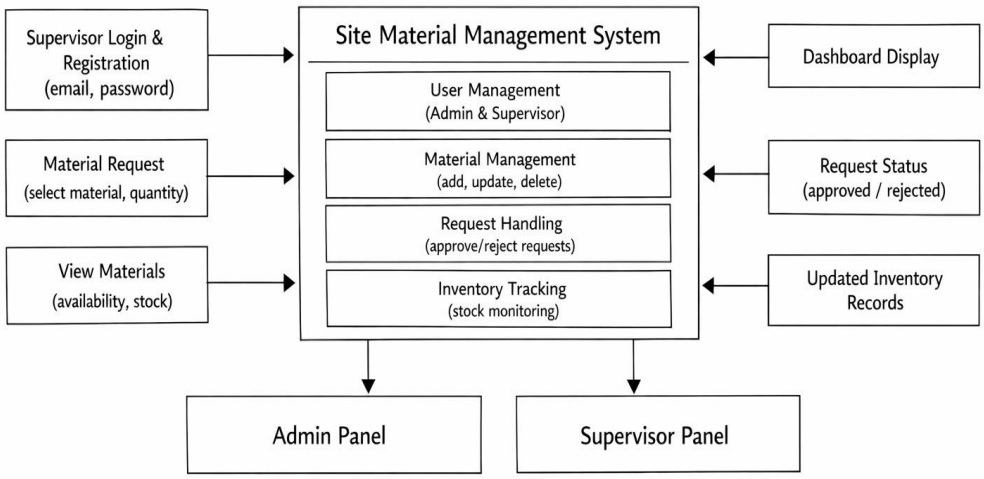

Figure 4.2: Proposed System For Site Management System

Fig. 4.2 illustrates how user inputs such as material details, stock entries, issue records, and usage data are processed by the system. The system interacts with the database and different modules to generate meaningful outputs like stock availability, material usage reports, and alerts for low stock. This helps in improving material tracking, reducing wastage, and ensuring efficient site operations

4.4.1 Material Management

This section explains how the system manages material details and categories.

- Stores material details such as name, type, quantity, and unit
- Allows adding, updating, and categorizing materials
- Maintains structured material records for easy access

4.4.2 Stock Management

This section describes how the system handles stock tracking and availability

- Records material stock entries and updates
- Tracks available and used quantities
- Provides real-time stock status

- Prevents stock shortages through proper monitoring

4.4.3 Payment & Cost Management

In this section, we will cover how the system manages material costs and payment records.

- Calculates total cost based on material quantity and rate
- Records payment details for purchased materials
- Tracks pending and completed payments
- Maintains cost history for reporting

In short, the system ensures accurate and transparent cost and payment tracking.

4.4.4 Material Issue & Return Management

In this section, we will explain how the system manages material issue and return processes.

- Records material issue details such as quantity and date
- Tracks material usage at the site
- Handles return of unused materials (if applicable)
- Identifies material wastage or damage

4.4.5 Inventory Monitoring

In this section, we will describe how the system tracks material availability and updates stock.

- Tracks availability of materials in real time
- Updates stock after issue and new purchase
- Maintains accurate stock records
- Alerts for low stock levels

4.5 User Management

In this section, we will cover how the system manages user access and ensures secure login.

- Provides role-based access control for Admin and Staff
- Ensures secure authentication using username and password
- Restricts access based on user roles

i. Input:

- Material details (Name, Type, Quantity, Unit)
- Stock entry and purchase details
- Material issue and usage data
- User login credentials (Username & Password)
- User role selection (Admin / Staff)

ii. System:

- Validates input data (correct format, required fields)
- Authenticates user login credentials
- Manages material records and categories
- Processes stock updates and material issue
- Tracks material usage and inventory status

iii. Database:

- Stores material details (Material ID, Name, Quantity, Unit)
- Maintains stock and transaction records
- Stores user details and roles (Admin / Staff)
- Tracks material usage and history

iv. Output:

- Material added / updated confirmation
- Stock availability status
- Material issue confirmation
- Low stock alerts
- Reports on material usage and inventory

4.6 Technology use to develop project:

- **Programming Language(s):** JavaScript
- **Frontend:** HTML, CSS, Bootstrap, React.js (using Vite.js)
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Development Tools**: GitHub (Version Control), Postman (API Testing),

Visual Studio Code

- **Operating System:** Windows 11

CHAPTER 5

Details of Designs, Working and Processes

This chapter provides a detailed explanation of the system design, workflow, and operational processes of the Site Material Management System. It describes how different modules interact with each other and how data flows through the system. The chapter includes system flow diagrams, module-wise operations, database design, and end-to-end working process.

5.1 System Flow Diagram

The Site Material Management System manages complete material operations from entry to usage tracking and inventory updates. It reduces manual errors by maintaining data in one centralized system and by updating records in real time. Every action performed by user is reflected across all connected modules, ensuring consistency and accuracy.

The system serves two main user roles along with a central control system. Each user interacts with the system in a defined and controlled manner. All operations are supported by a database system that stores and retrieves information related to materials, stock, usage, and reports. The system has three user-facing roles and one central control role:

- Admin — system configuration and monitoring
- Staff (Site Supervisor / Store Manager) — daily material operations (material entry, issue, stock update, tracking)
- Database system— real-time data storage, processing and retrieval

  The diagram below shows how each actor interacts with the central Site Material Management System and what outputs are produced:

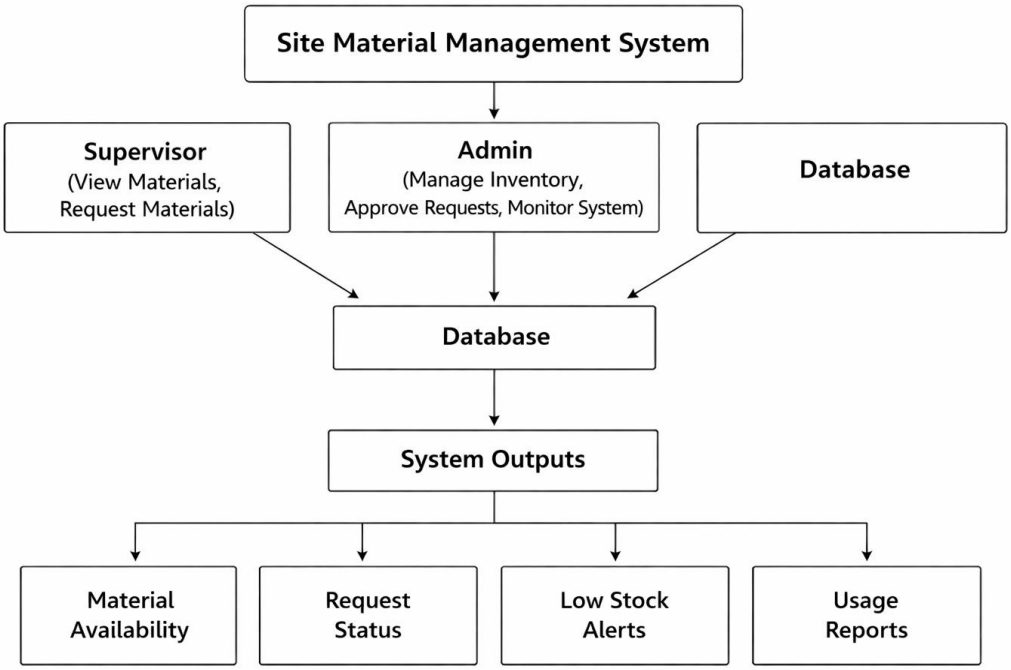

Figure 5.1: System Flow Diagram — Actors, System Core, and Outputs

5.2 Admin Module

The Admin Module acts as the central control unit of the system. It is responsible for managing core system configurations and supervising all activities related to material management. The Admin has full access to system data, including materials, stock levels, user activities, and reports.

This module ensures that the system operates smoothly, maintains data accuracy, and follows the defined operational structure. The Admin plays a key role in organizing materials, assigning responsibilities, and monitoring overall system performance. Admin responsibilities mainly include managing material records, controlling user access, and analyzing reports to support decision-making. A well-functioning Admin Module is essential for maintaining system efficiency and reliability.

Key features:

- Manage users and assign role-based permissions
- Add, update, and organize material categories and materials
- Monitor material stock, usage, and inventory updates

- Track overall system activities and operations

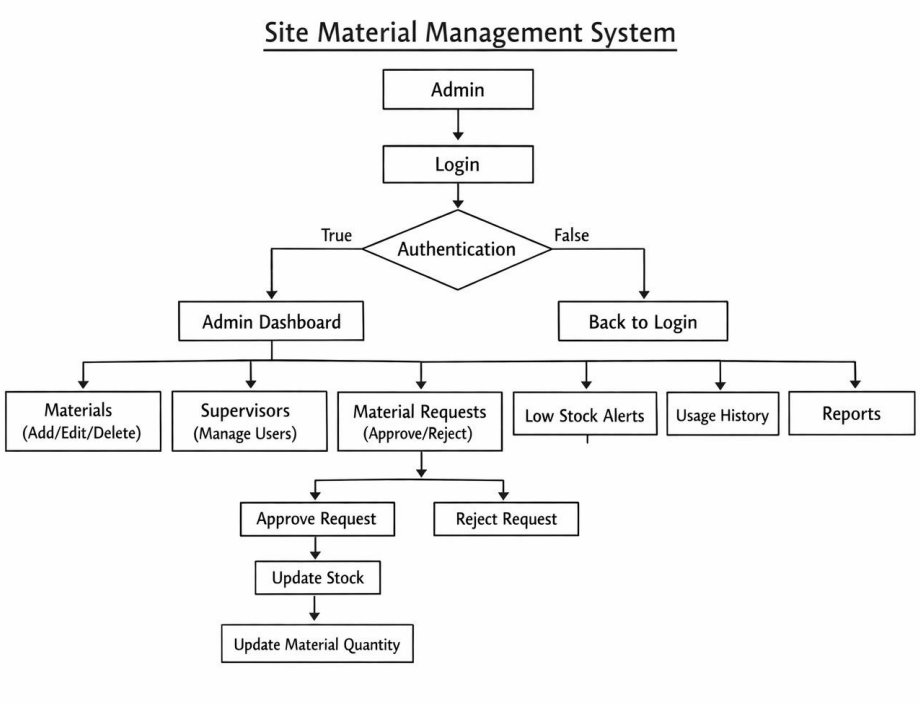

Figure 5.2: Admin Module Flow — Login to Report Generation and Logout

5.3 Staff Module

The Staff Module is responsible for managing all daily material handling activities at the site. Staff members are the main users who interact with the system to manage materials, track usage, and maintain accurate stock records. This module supports the complete lifecycle of material operations, starting from checking availability to issuing materials and updating inventory.

Staff do not have access to system configuration, but they are provided with full operational access to material handling functions. The module is designed to be simple, fast, and efficient, allowing staff to complete tasks quickly and accurately

Key features:

- Create and update material issue records
- Record material usage at the site
- Handle return of unused materials (if applicable)

- Update stock and material status after usage
- Record cost details and transaction updates

The following flow diagram represents the complete Staff Module workflow. It shows how staff check material availability, issue materials, record usage, and update stock. It also includes decision points for handling material damage or excess usage. This workflow ensures proper tracking, reduces errors, and improves efficiency in site material management.

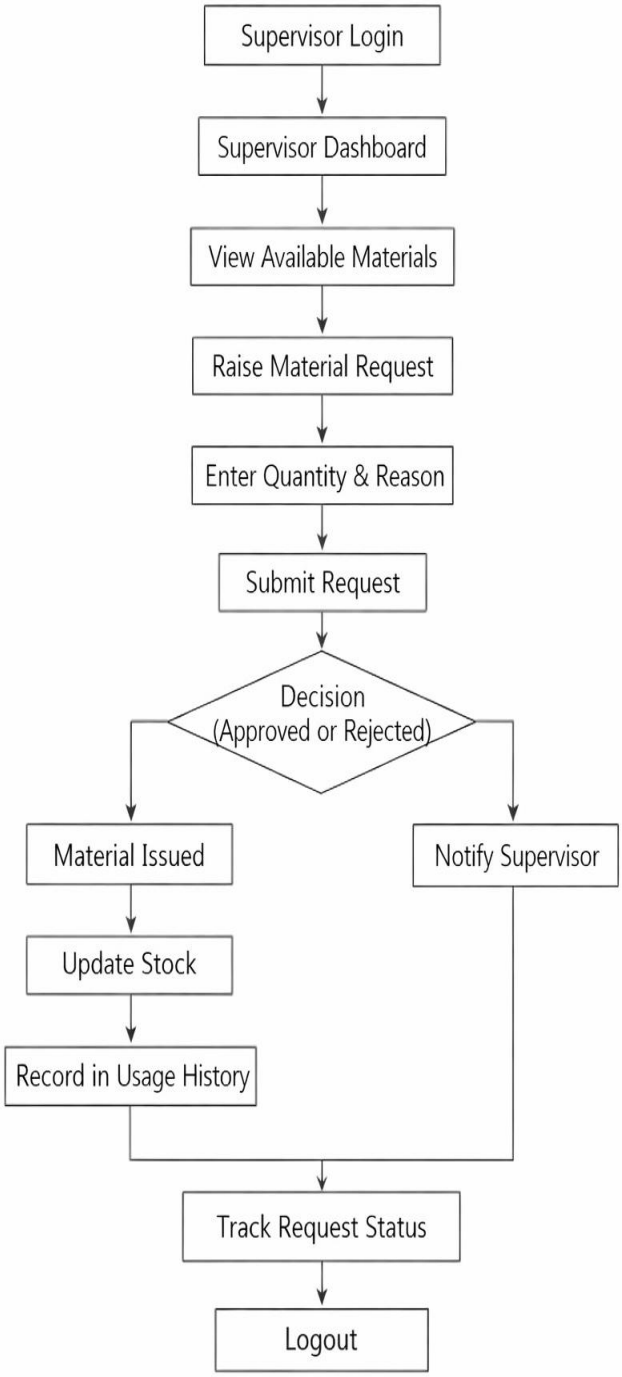

Figure 5.3 : Staff Module Flow

5.4 Use Case Diagram

The Use Case Diagram represents the interaction between system users and the functionalities available in the Site Material Management System. It provides a high-level overview of how different users access and use the system features based on their roles. The system includes two primary actors Admin and Staff each having specific responsibilities and access rights within the system.

The Use Case Diagram helps in clearly understanding user roles, system boundaries, and how different functionalities are accessed within the system.

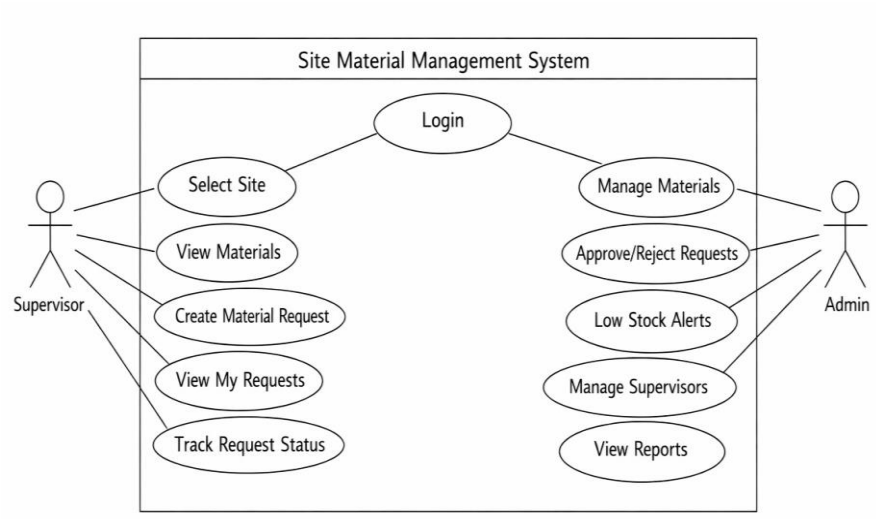

Figure 5.5: Use Case Diagram — System Actors and Role-wise Use Cases

5.5 Database Schema

The database acts as the backbone of the Site Material Management System, storing all operational data in a structured format and supporting the complete workflow—from user authentication and material entry to stock tracking, usage monitoring, and report generation..

For instance, each material transaction is linked to a specific material and user, enabling complete traceability of material usage. Similarly, stock records are

automatically updated based on material entry and issue activities.

The following table describes all database entities, their purpose, and the key fields they contain:

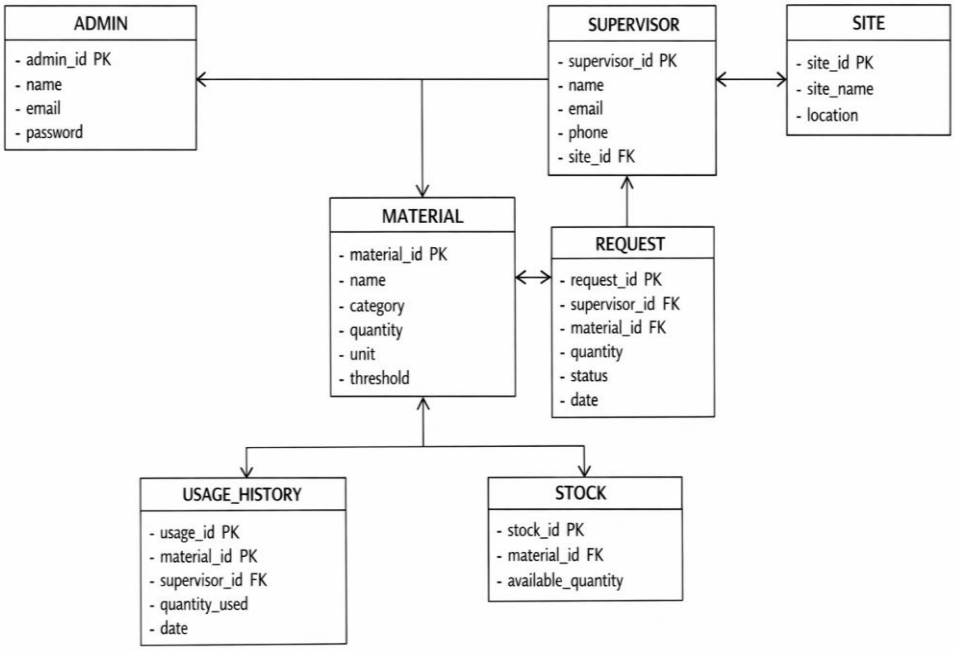

Figure 5.6 : Database ER Diagram Entity Relationships

List of Database Tables / Collections:-

The system uses the following eight database collections. Each collection is focused on a specific operational area and is linked to related collections through reference fields

The end-to-end working process of the Site Material Management System explains how material-related operations are carried out from initial system setup to final reporting. It connects all modules and user roles into a single, well-defined workflow.

The complete working process follows these steps:

1. Admin sets up users, material categories, and material details.
2. Staff checks material availability and records material requirements.
3. Materials are issued for site work and stock is updated accordingly.
4. Material usage is recorded during or after site operations.
5. Any remaining, damaged, or wasted materials are identified.

CHAPTER 6

RESULT AND APPLICATION

This chapter presents the results of the developed Site Material Management System through major web pages and their functional descriptions. The screenshots demonstrate how different users interact with the system for material management operations such as material entry, stock tracking, issue management, and report monitoring.

The public-facing pages of the system are accessible before login. These pages introduce the system, guide new users through registration, and allow existing users to log in securely. They act as the entry point for system users such as Admin and Staff

6.1 Home Page

Figure 6.1 shows the home page of the Site Material Management System. This page serves as the entry point of the application and provides a clear overview of the system's purpose and functionality..

Additionally, the home page plays an important role in guiding new users by providing information about the system's functionality and benefits. It ensures smooth interaction by offering a simple and organized layout, making it easy for both administrators and supervisors to proceed with their tasks.

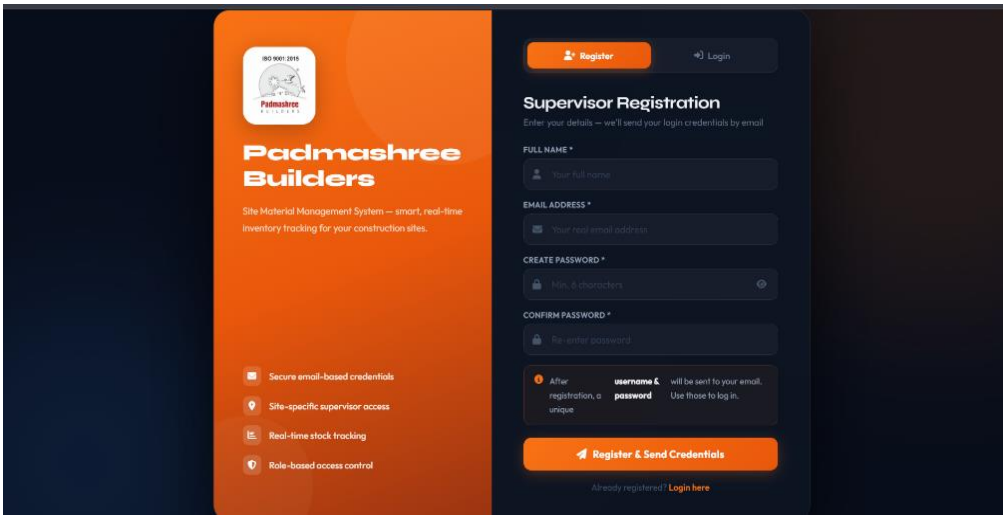

Figure 6.1: Home Page — System Overview and Navigation.

Main Outcomes :

User-friendly Interface: The home page provides a clean, simple, and well-structured layout, making it easy for users to understand the system and interact with different features without confusion.

Easy Navigation: Users can quickly access important sections such as material management, login, and registration through clearly visible navigation options, improving overall usability.

System Introduction: The page clearly explains the purpose of the system and highlights its benefits in managing site materials efficiently and reducing manual effort.

Improved Accessibility: The design ensures that both new and existing users can easily explore the system and access required functionalities without difficulty

6.2 Login Page:

The Login Page of the Site Material Management System allows users to securely access the system based on their roles. It is an essential component that ensures only authorized users can enter the system and perform operations.

Overall, the login page plays a crucial role in controlling access, maintaining data security, and ensuring that users can efficiently interact with the system based on their permissions.

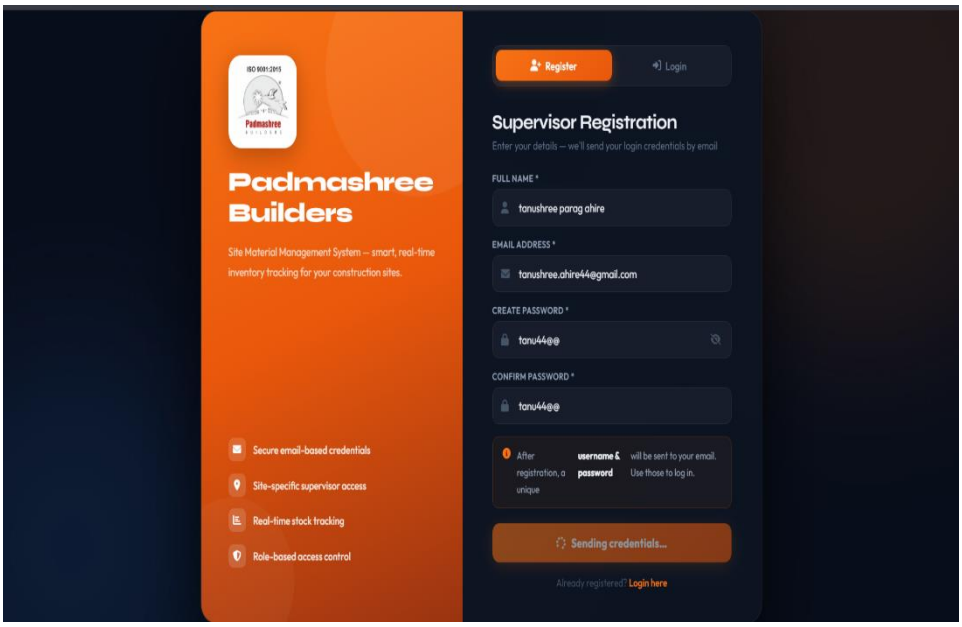

Figure 6.2: Login Page – Secure Credential-Based Authentication

Table 6.1: Registration Page — Supervisor Account Creation

Fields	**Validation Applied**
Full Name	Must not be empty; accepts alphabetic characters
Email Address	Must be in valid email format and unique
Create Password	Must not be empty; minimum length required
Confirm Password	Must match the entered password
Empty Fields Validation	All fields are mandatory
Register Button	Enabled only when all fields are correctly filled

Main Outcomes:

Secure Authentication: Ensures that only authorized users can access the system using valid credentials

Role-Based Access: Redirects users (Admin/Supervisor) to their respective dashboards

Data Protection: Prevents unauthorized access and protects sensitive system data

Input Validation: Restricts empty fields and incorrect login attempts

Error Handling: Displays appropriate error messages for invalid credentials

6.3 Check Email Page

Figure 6.3 shows the Check Email page displayed after the supervisor successfully completes the registration process in the Site Material Management System. This page acts as an important step in the user onboarding process by informing the user about the next actions required to access the system.

After registration, the system automatically generates unique login credentials and sends them to the user's registered email address. The Check Email page clearly notifies the user about this process and instructs them to check their email inbox to retrieve their username and password. This ensures that only users with access to the registered email can log in, thereby enhancing system security.

The page also provides clear guidance to avoid confusion, especially for new users. It may include options such as redirecting to the login page after checking the email or retrying the process if the email is not received. By providing proper instructions and feedback, this page ensures a smooth transition from registration to login.

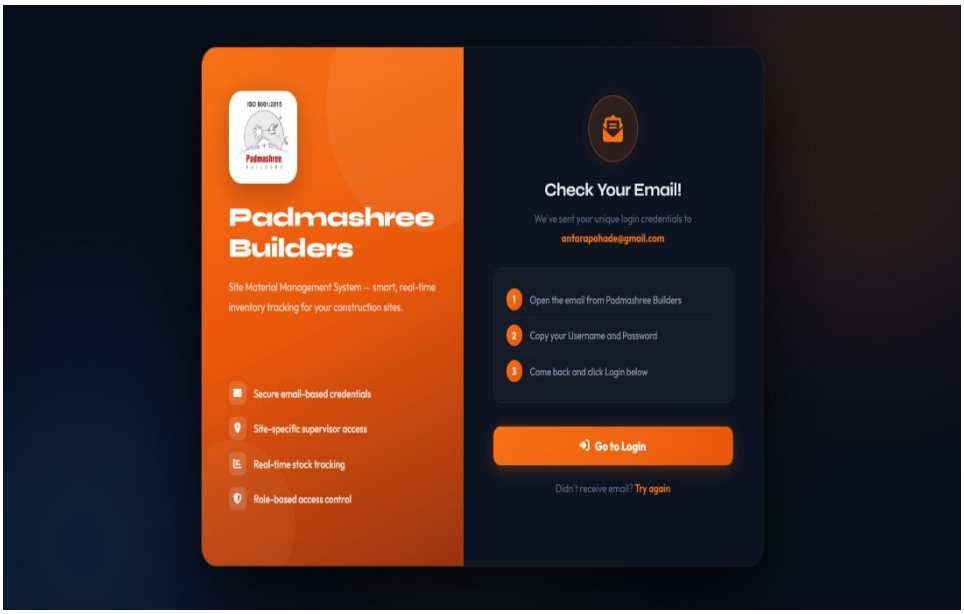

Figure 6.2: Check Email Page Check Email Page Credential Notification after Signup

Table 6.2: Check Email Page Validations

Fields	**Validation Applied**
Email Display	Shows the registered email address where credentials are sent
Instruction Steps	Provides step-by-step guidance to access login credentials
Go to Login Button	Redirects user to login page
Resend Option (Try Again)	Allows user to request email again if not received
User Guidance Message	Ensures user understands next steps after registration

Main Outcomes

Guided User Flow: Clearly directs users on the next step after registration (checking email)

Secure Credential Delivery: Ensures login credentials are sent only to the registered email

Improved User Awareness: Notifies users about successful registration and credential dispatch

Reduced Confusion: Provides clear instructions for accessing the system

Enhanced Security: Ensures only users with email access can log in

Smooth Transition: Connects registration process to login seamlessly

6.4 Select Site Page

Figure 6.4 shows the Site Selection page displayed after the supervisor logs in using the unique credentials received via email. This page plays an important role in ensuring that the user accesses the correct working environment within the system. The page is designed with a simple and user-friendly interface, allowing easy selection of the assigned site. It also prevents unauthorized access by restricting users to only those sites that are assigned to them. Once the site is selected, the user is redirected to the dashboard, where they can continue their tasks efficiently.

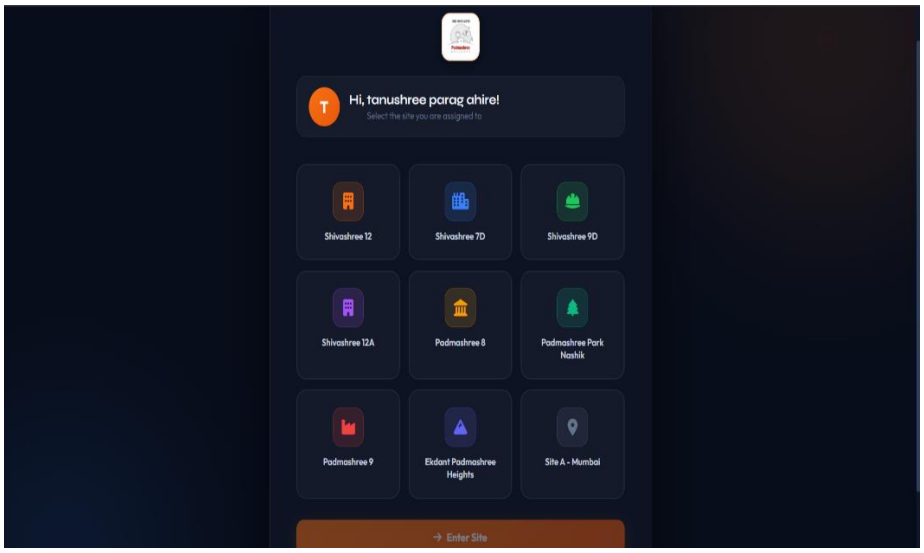

Figure 6.4: Selection Site Page

Table 6.3: Selection Site

Fields	Validation Applied
User Greeting	Displays logged-in user name for personalization
Site List	Shows only the sites assigned to the supervisor
Site Selection	User must select one site to proceed
Enter Site Button	Enabled only after selecting a site
Access Control	Prevents access without selecting a valid site
Navigation	Redirects user to dashboard after site selection

Main Outcomes

Site-Based Access Control: Ensures users access only the sites assigned to them

Accurate Data Management: All operations are performed within the selected site context

Improved Organization: Separates data and activities based on different construction sites

User Guidance: Directs users clearly after login to choose their working site

Enhanced Security: Prevents unauthorized access to other sites

Smooth Workflow: Provides seamless transition from login to dashboard

6.5 Supervisor Dashboard

Figure 6.5 shows the Supervisor Dashboard of the system. This page is displayed after the supervisor successfully logs in and selects a site. It acts as the main interface for the supervisor, providing a clear and organized overview of all important activities related to material management.

The dashboard presents key information such as available materials, total requests, pending requests, and recent activities. This helps the supervisor quickly understand the current status of materials and ongoing requests without navigating through multiple pages. It also improves decision-making by providing real-time updates from the system.

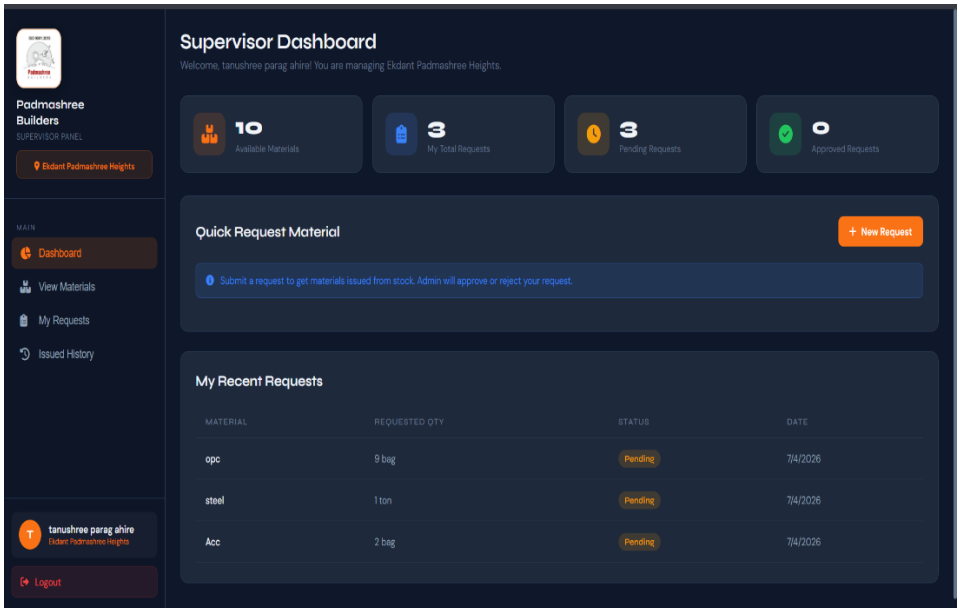

Figure 6.5: Supervisor Dashboard Validations

Table 6.4: Supervisor Dashboard Validations

Fields	Validation Applied
User Information	Displays logged-in supervisor name and selected site
Available Materials Count	Shows total number of materials available in the system
Total Requests	Displays number of requests raised by the supervisor
Pending Requests	Shows requests that are awaiting approval
Approved Requests	Displays successfully approved requests
Quick Reques Section	Allows user to initiate a new material request
New Reques Button	Redirects to request form page
Recent Requests Table	Displays recently requested materials with quantity, status, and date
Logout Option	Allows secure exit from the system

Main Outcomes

- Centralized Monitoring: Provides a complete overview of all system activities in one place
- Quick Decision-Making: Displays key metrics for faster& informed decisions
- Low Stock Awareness: Highlights materials that require

6.6 Available Materials Page.

Figure 6.6 shows the Available Materials page of the system. This page allows the supervisor to view the current stock of materials available at the selected site. It serves as an important interface for monitoring inventory and ensuring that sufficient materials are available for ongoing construction activities.

The page displays detailed information such as material name, category, available quantity, unit of measurement, and stock status. This organized presentation helps the supervisor quickly understand the availability of each material and identify items that may be running low. The stock status further assists in highlighting whether a material is sufficiently available or needs attention.

In addition to viewing material details, the supervisor can directly raise a material request from this page. This feature improves efficiency by allowing users to request required materials without navigating to a separate section. It ensures a smooth workflow by linking material availability with the request process.

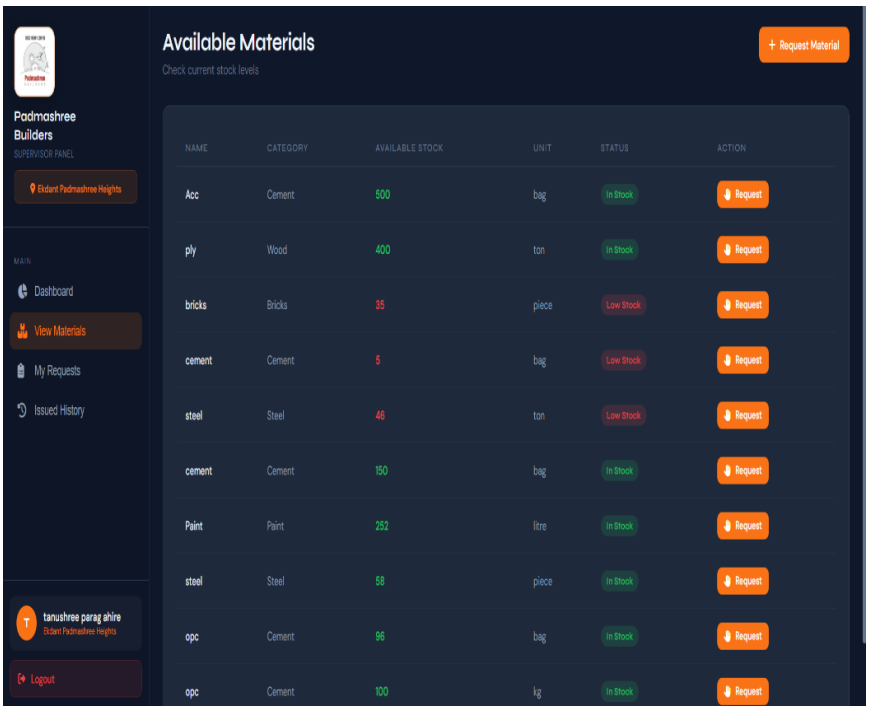

Figure 6.6: Available Materials Page Material Stock Overview

Table 6. 5: Available Materials Page Material Stock Overview Validations

Fields	Validation Applied
Material Name	Displays the name of the material
Category	Shows the category of each material
Available Stock	Displays current quantity; dynamically updated from database
Unit	Shows measurement unit (e.g., bag, ton, piece, litre)
Status Indicator	Automatically shows "In Stock" or "Low Stock" based on quantity
Request Button	Allows user to request selected material
Request Action	Redirects to request with selected material
Low Stock Highlight	Highlights materials with low quantity for quick attention
Data Accuracy	Data fetched in real-time from backend
Navigation Menu	Allows switching between dashboard, materials, requests, and history

Main Outcomes

- Real-Time Stock Visibility: Provides up-to-date information on available materials
- Better Decision-Making: Helps supervisors identify which materials are available or low

- Efficient Request Process: Allows direct material request from the same page
- Improved Inventory Control: Ensures proper monitoring of stock levels
- Reduced Delays: Helps avoid work interruptions by checking availability

6.7 My Bookings Page

Figure 6.7 shows the My Requests page of the system. This page allows the supervisor to track all the material requests raised by them. It serves as an important module for monitoring the status and progress of requested materials within the system.

The page displays detailed information such as material name, category, requested quantity, request status, admin remarks, and request date. This structured view helps the supervisor easily understand the current state of each request. The status field clearly indicates whether a request is pending, approved, or processed, allowing better tracking and planning.

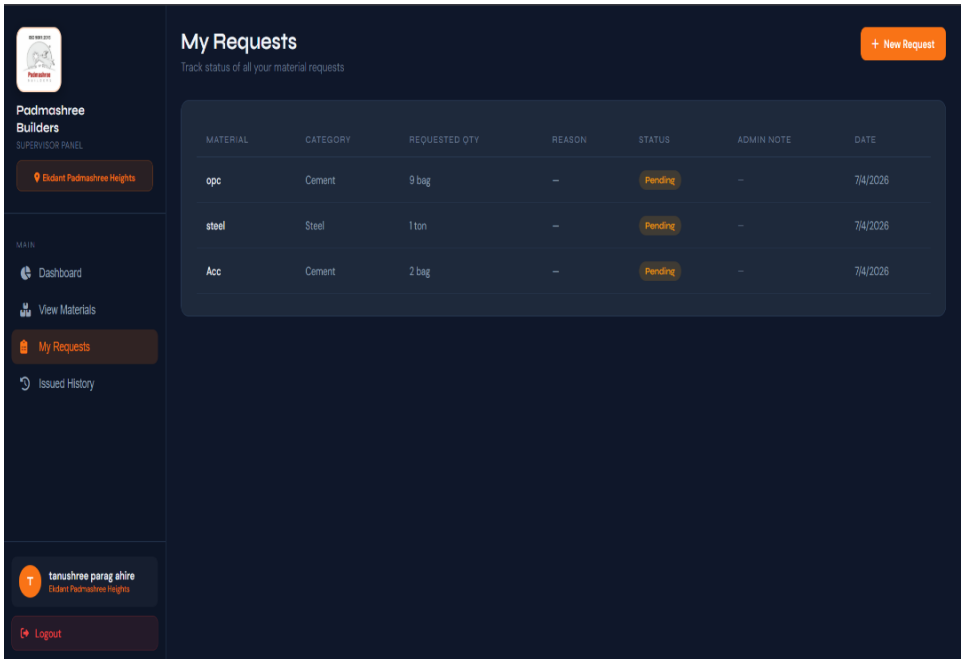

Figure 6.7: My Requests Page Request Tracking and Status Monitoring

Table 6.6 My Requests Page Validations

Fields	**Validation Applied**
Material Name	Displays the name of the requested material
Category	Shows the category of the material
Requested Quantity	Displays quantity requested by the supervisor
Reason	Optional field to specify purpose of request
Status	Shows current status (e.g., Pending, Approved)
Admin Note	Displays remarks or feedback from admin (if any)
Date	Shows request submission date
New Request Button	Allows user to create a new material request
Data Accuracy	Data fetched dynamically from backend database
Status Indicator	Highlights request status for easy tracking

Main Outcomes

Request Tracking: Track all requests in one place.

Improved Transparency: Provides clear info including admin remarks and request details

Better Planning: Helps supervisors plan work based on request status

Efficient Communication: Reduces the need for direct follow-up admin.

6.8 Request Material Page

Figure 6.8 shows the Request Material form displayed when the supervisor clicks on the "New Request" option. This form is used to raise a request for required materials in a structured and efficient manner within the system.

The form allows the user to select a material from the available list, enter the required quantity, and optionally provide a reason for the request. Displaying available stock while selecting the material helps the supervisor make informed decisions and avoid requesting unavailable quantities.

The system also applies proper validation mechanisms before submitting the request. It ensures that all required fields, such as material selection and quantity, are correctly filled. Invalid or empty inputs are restricted, preventing incorrect data from being submitted to the system. Once validated, the request is stored in the database and forwarded to the admin for review and approval.

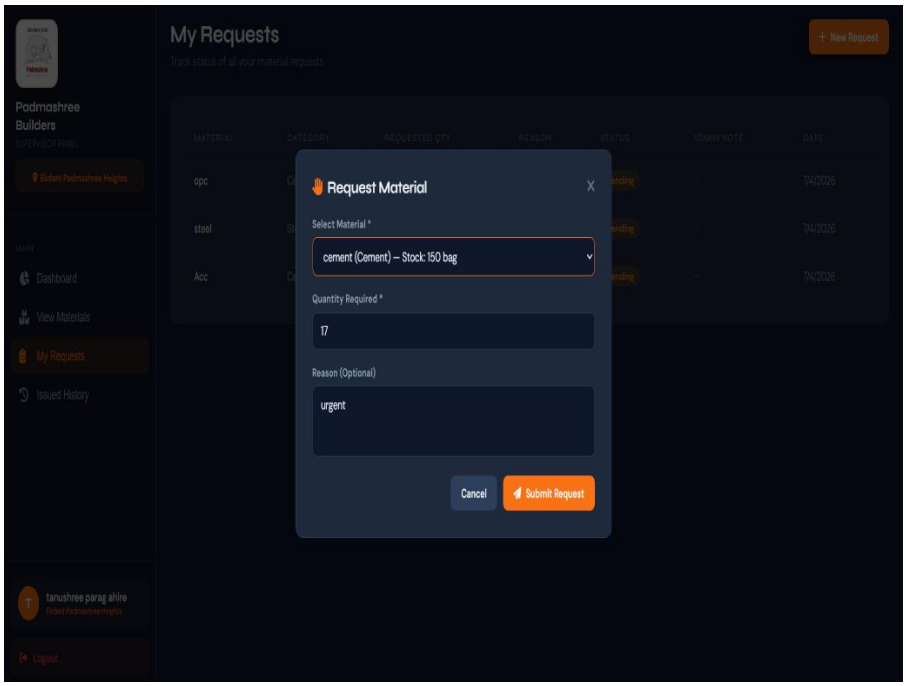

Figure 6.8: Request Material Form Material Request Submission

Table 6.7: Request Material Form Material Request Submission

Fields	**Validation Applied**
Select Material	User must select a material from the dropdown list
Stock Display	Shows available stock for selected material
Quantity Required	Must be a positive number and should not be empty
Quantity Validation	Cannot exceed available stock (if implemented)
Reason	Optional field for specifying request purpose
Empty Fields Validation	Material and quantity fields are mandatory

Submit Request btn	Enabled only when required fields are fille correctly
Cancel Button	Closes the request form without saving
Data Submission	Sends request to backend and stores in database
User Feedback	Displays success message after requ submission

Main Outcomes:-

Easy Material Request Submission: Quickly submit material requests.

Improved Efficiency: Simplifies the request process through a user-friendly form

Real-Time Processing: Sends requests directly to the system for admin review

Reduced Errors: Prevents invalid or incomplete submission.

6.9 Admin Login Page

Figure 6.9 shows the Admin Login page of the system. This page is specifically designed for authorized administrators to securely access the Site Material Management System and manage overall operations.

The admin is required to enter valid login credentials, including email address and password, which are verified against the stored records in the database. This authentication process ensures that only authorized users can gain access to the admin panel. Once the credentials are successfully verified, the admin is redirected to the dashboard, where they can manage materials, supervise requests, monitor stock levels, and control user activities.

The login page also includes important validation mechanisms to

maintain system security. It prevents submission of empty fields and displays appropriate error messages when incorrect credentials are entered.

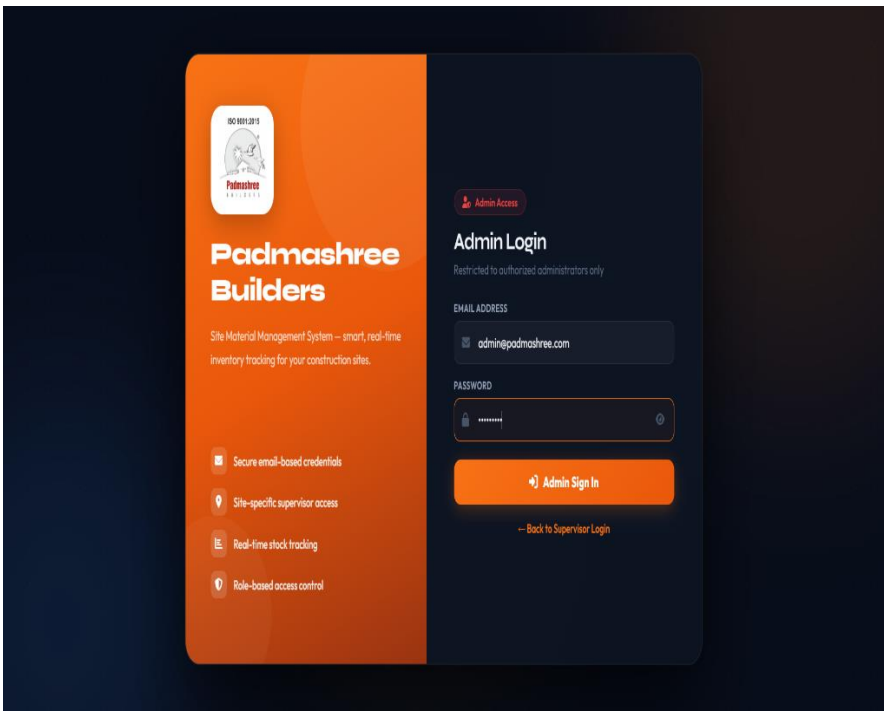

Fig. 6.9: Booking Payment Page — Payment Records and Settlement Status

Table 6.8: Admin Login Page Validations

Admin Dashboard

Fields	**Validation Applied**
Email Address	Must be in valid email format (e.g., admin@domain.com)
Password	Must not be empty and should match registered admin credentials

Empty Fields Validation	Both email and password fields are mandatory
Invalid Credentials	Displays error message if login details are incorrect
Admin Sign In Button	Enabled only when all required fields are filled
Role-Based Access	Only authorized admin users can access this panel
Password Visibility Toggle	Allows user to show/hide password
Navigation (Back to Supervisor Login)	Redirects user to supervisor login page

Figure 6.10 Admin Dashboard

Figure 6.10 shows the Admin Dashboard of the system. This page provides a comprehensive overview of all activities across different construction sites within the Site Material Management System.

Overall, the Admin Dashboard enhances system efficiency by providing centralized information, supporting quick decision-making, and ensuring smooth management of materials, users, and requests across all sites.

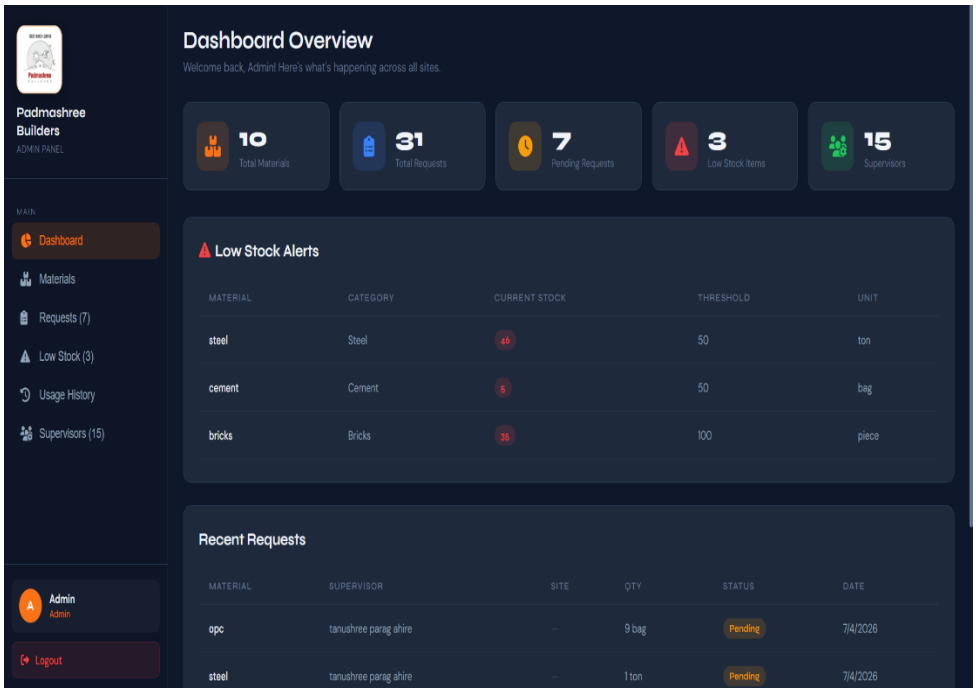

Figure 6.10: Admin Dashboard System Overview and Monitoring

Table 6.9: Admin Dashboard Validations – System Overview & Management

Fields	Validation Applied
Total Materials	Displays total number of materials available in the system
Total Requests	Shows total number of material requests raised
Pending Requests	Displays requests awaiting approval
Low Stock Items	Highlights materials with stock below threshold
Low Stock Alerts Table	Displays materials with low quantity along with threshold values

Recent Requests Table	Shows latest material requests with supervisor name, quantity, and status
Status Indicator	Displays request status (e.g., Pending, Approved)

Main Outcomes

- **Centralized Panel:** All activities in one place
- **Real-Time Monitoring:** Shows updated data
- **Quick Decisions:** Supports fast decisions
- **Low Stock Alert:** Identifies low materials
- **Request Management:** Tracks requests
- **Efficiency:** Improves overall system

6.11 Admin Dashboard (Low Stock Alerts Validations)

Figure 6.11 shows the detailed section of the Admin Dashboard, highlighting Low Stock Alerts and Recent Requests. This section plays a crucial role in helping the admin monitor critical inventory levels and track ongoing activities across different construction sites.

The Low Stock Alerts section identifies materials whose stock levels have fallen below the predefined threshold. It provides a clear indication of items that require immediate attention, enabling the admin to take timely actions such as restocking.

The Recent Requests section displays the latest material requests submitted by supervisors. It includes important details such as material name, requested quantity, site, and request status. This allows the admin to quickly review and respond to new requests, improving communication and workflow efficiency between supervisors and administrators.

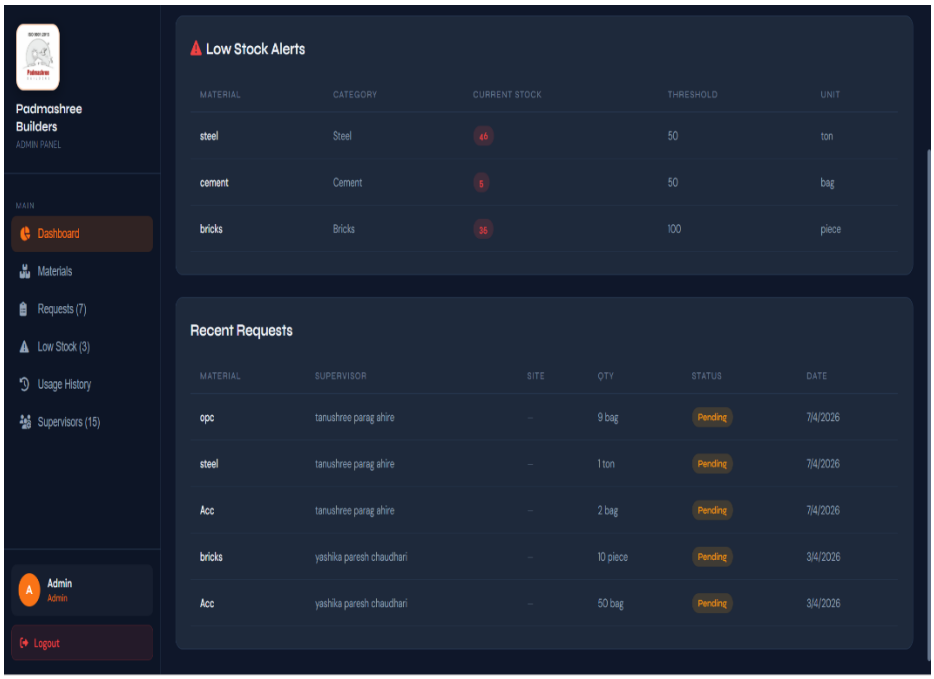

Figure 6.11: Low Stock Alerts and Recent Requests

Table 6.10: Low Stock Alerts Recent Requests Validations

Fields	**Validation Applied**
Material Name	Displays name of material with low stock
Category	Shows material category
Current Stock	Displays current available quantity
Threshold	Predefined minimum stock level
Low Stock Detection	Automatically flags materials below threshold
Unit	Displays unit of measurement

Highlight Indicator	Low stock values are visually highlighted
Data Accuracy	Data fetched dynamically from database

Table 6.10: Low Stock Alerts Recent Requests Validations

Main Outcomes

- **Efficient Request Management:** Enables admin to review and handle all material requests in one place
- **Clear Status Tracking:** Shows request status such as pending, approved, or rejected
- **Quick Decision-Making:** Allows admin to approve or reject requests instantly
- **Improved Resource Allocation:** Ensures materials are distributed based on actual need.

6.12 Materials Page

Figure 6.12 shows the Materials page of the system. This page allows the administrator to efficiently manage all construction materials within the Site Material Management System. It acts as a central interface for handling inventory-related operations across different sites.

The page displays detailed information such as material name, category, stock quantity, unit of measurement, price per unit, threshold value, and stock status of materials. The inclusion of threshold values enables the system to identify low stock items, ensuring timely restocking and avoiding shortages.In addition to viewing material details, the admin can perform key operations such as adding new materials, editing existing records, and deleting materials when required. These functionalities ensure that the

inventory remains accurate, up-to-date, and well-organized

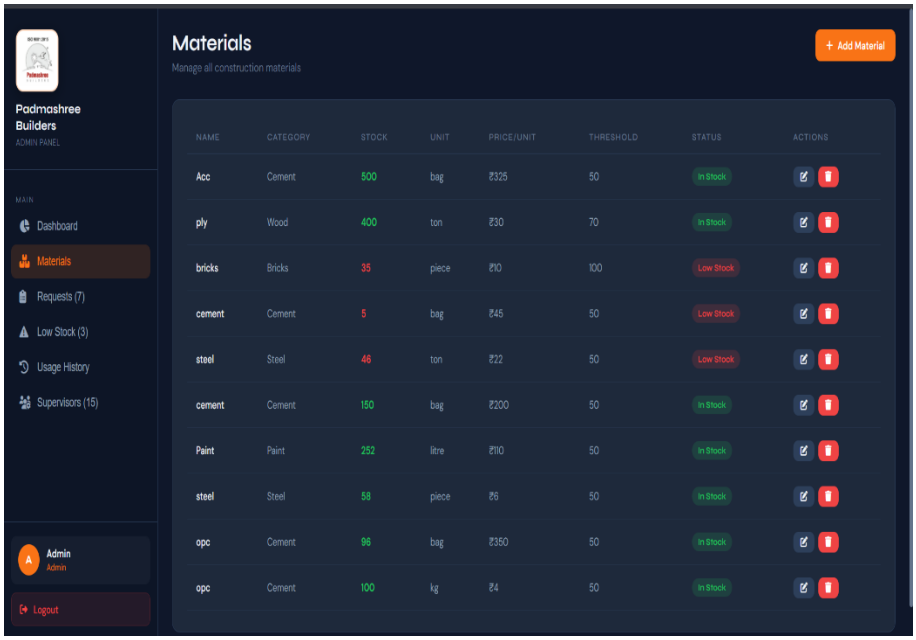

Figure 6.12: Admin Materials Page Validations

Table 6.11: Admin Materials Page Validations

Fields	**Validation Applied**
Material Name	Displays name of material with low stock
Category	Shows material category
Stock Quantity	Must be a numeric value and automatically updated
Price per Unit	Must be a valid numeric value (₹)
Threshold Value	Defines minimum stock level for alerts
Unit	Displays unit of measurement (e.g., ton, bag, piece)

Stock Status	Automatically shows “In Stock” or “Low Stock”
Low Stock Detection	Highlights materials below threshold in red
Add Material Button	Allows admin to add new material
Edit Action	Enables modification of existing material details
Delete Action	Allows deletion with confirmation
Data Accuracy	Data is fetched and updated in real-time from database
Navigation Menu	Allows access to dashboard, requests, stock alerts, etc

Main outcomes:

Efficient Request Management: Enables admin to review and handle all material requests in one place

Clear Status Tracking: Shows request status such as pending, approved, or rejected

Quick Decision-Making: Allows admin to approve or reject requests instantly

Improved Resource Allocation: Ensures materials are distributed based on actual needs

Enhanced Transparency: Displays complete request details including reason and date

Better Coordination: Strengthens communication.

6.13 Add New Material Form

Figure 6.13 shows the “Add New Material” form used by the admin

to add new construction materials into the system. This form is an important component of the Site Material Management System, as it allows the administrator to maintain and update the inventory efficiently.

The admin is required to enter essential details such as material name, category, quantity, unit of measurement, price per unit, and threshold value. These details help in accurately defining each material and ensuring proper tracking within the system.

The form is designed with proper validation mechanisms to ensure that all necessary fields are filled correctly before submission. It prevents invalid or incomplete data from being entered into the system, thereby maintaining data accuracy and consistency.

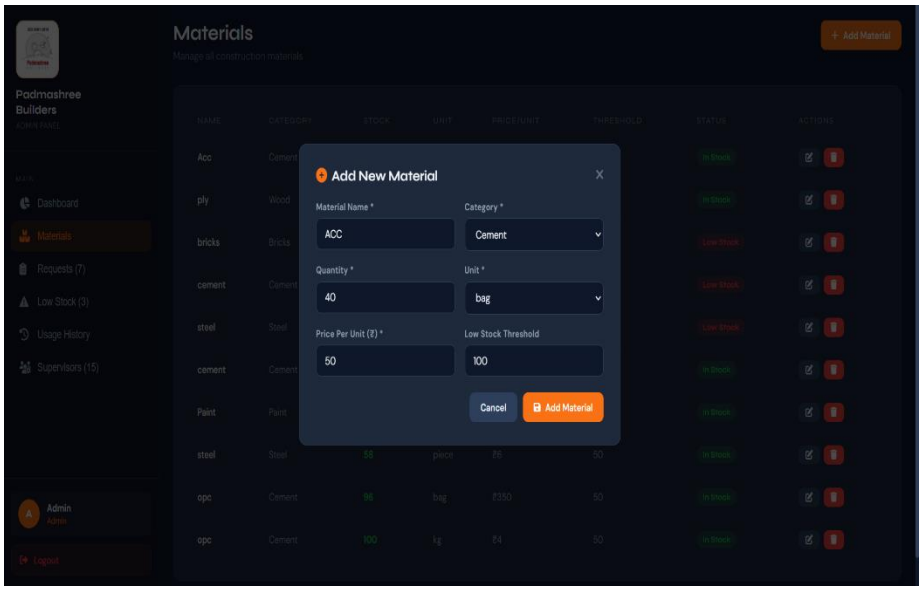

Figure 6.12: Add New Material Form Material Entry Interface

Table 6.13: Add New Material Form Material Entry Interface Validations

Fields	Validation Applied

Material Name	Displays name of material with low stock
Category	Shows material category
Quantity	Must be a positive numeric value
Price Per Unit	Must be a valid numeric value (₹)
Low Stock Threshold	Must be a numeric value to define minimum stock level
Unit	Displays unit of measurement (e.g., ton, bag, piece)
Empty Fields Validation	All fields are mandatory and cannot be left blank
Add Material Button	Enabled only when all inputs are valid
Cancel Button	Closes the form without saving data

Main outcomes:

- **Efficient Inventory Management:** Allows admin to manage all materials in one place
- **Real-Time Stock Monitoring:** Provides updated information on material availability
- **Low Stock Identification:** Helps detect materials below threshold levels
- **Easy Data Management:** Supports adding, editing, and deleting

materials

- **Improved Accuracy:** Ensures consistent and error-free material records
- **Better Decision-Making:** Assists admin in planning procurement and stock control

6.14 Material Requests Page

Figure 6.14 shows the Material Requests page of the system. This page allows the admin to review and manage all material requests submitted by supervisors across different construction sites. It plays a key role in ensuring proper control and allocation of materials within the system. The admin can directly approve or reject requests from this interface, making the process efficient and time-saving. Once an action is taken, the request status is updated accordingly, ensuring transparency for both admin and supervisor. This helps maintain a smooth workflow and clear communication between users.

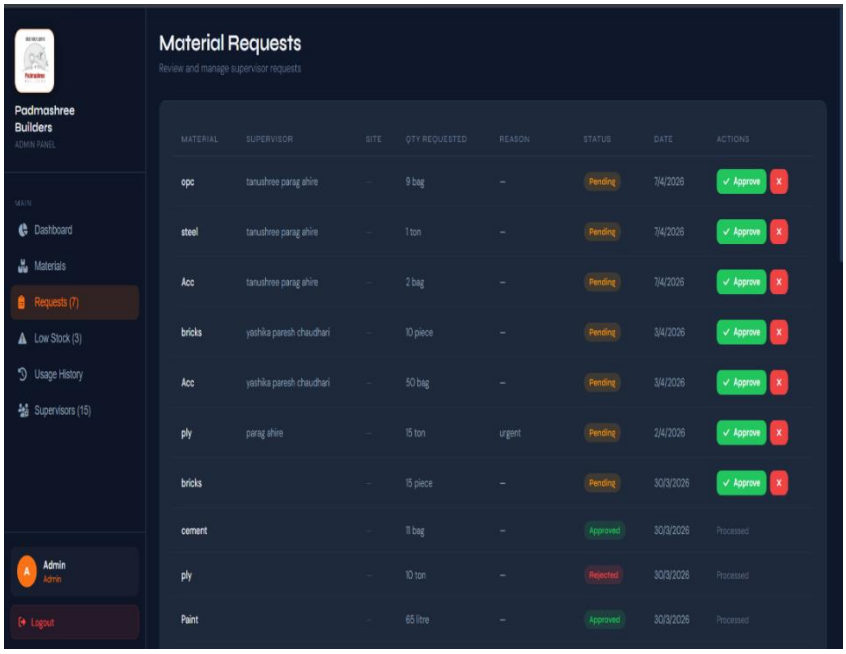

Figure 6.14: Material Requests Page Request Review and Approval

Table 6.13: Material Requests Page Request Review and Approval Validations

Fields	Validation Applied
Material Name	Displays the requested material
Supervisor Name	Shows the name of the supervisor who rais the request
Site	Displays associated site (if available)
Quantity Requested	Must be a valid numeric value
Reason	Optional field for request justification
Status	Displays request status
Date	Shows request submission date
Approve Button	Allows admin to approve pending requests
Status Update	Automatically updates status after admi action
Processed State	Disables actions once request approved/rejected

Main Outcomes

- **Centralized Request Handling**: Allows admin to manage all material **requests** in one place
- **Clear Status Tracking**: Displays request status (Pending, Approved, Rejected)
- **Quick Decision-Making**: Enables fast approval or rejection of requests

- **Efficient Material Allocation**: Ensures proper distribution of materials based on needs
- **Transparency**: Provides complete request details including reason.

6.15 Low Stock Alerts Page

Figure 6.15 shows the Low Stock Alerts page of the system. This page is designed to help the administrator monitor materials whose stock levels have fallen below the predefined threshold. It plays a crucial role in maintaining proper inventory levels and ensuring that construction activities are not interrupted due to material shortages.

The page displays a list of materials with low stock along with important details such as material name, category, current stock, threshold value, unit, and price per unit.

Additionally, the system provides an "Update Stock" option, which enables the admin to directly increase the stock quantity from the same page. This feature improves efficiency by reducing the need to navigate to different sections for stock updates. The automatic detection of low stock based on threshold value

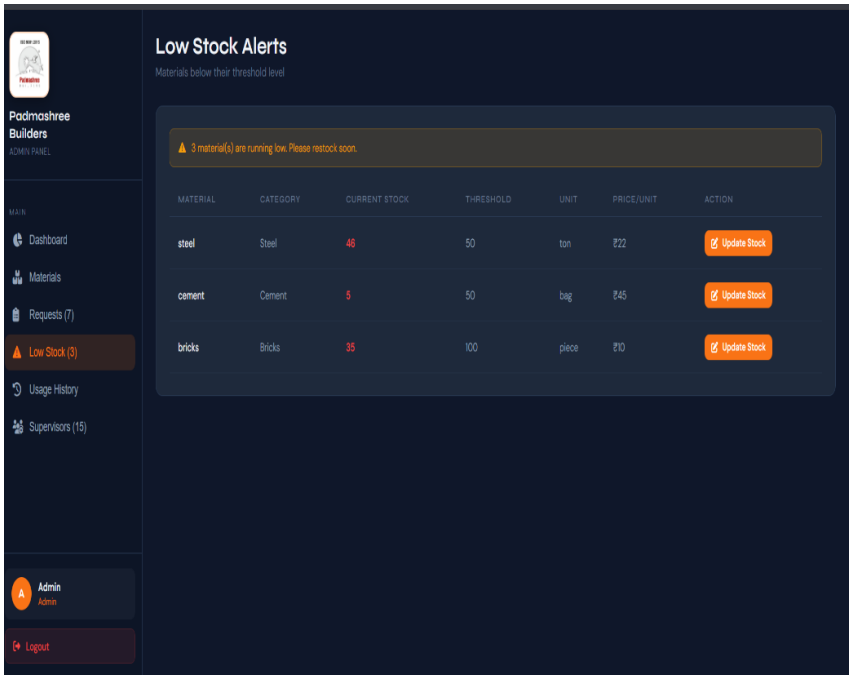

Figure 6.15: Low Stock Alerts Page — Low Stock Monitoring and Alerts

Table 6.14 Low Stock Alerts Page -- Low Stock Monitoring and Alerts Validation

Fields	Validation Applied
Alret Message	Displays warning (e.g., “3 materials are running low”)
Material Name	Shows materials with low stock
Category	Displays material category
Current Stock	Highlighted in red when below threshold
Threshold Value	Defines minimum stock level
Low Stock	Automatically identifies low stock items
Unit	Displays unit (ton, bag, piece, etc.)
Update Stock Button	Allows admin to update stock quantity

Main Outcomes

- **Early Detection of Low Stock**: Identifies materials that fall below the threshold level
- **Timely Alerts:** Provides warning messages for quick action
- **Prevents Shortages:** Helps avoid delays in construction due to insufficient materials
- **Quick Stock Update:** Allows admin to update stock directly

from the same page

- **Improved Inventory Control:** Ensures proper monitoring of material levels

- **Better Planning:** Supports timely procurement
- **6.16 Usage History Page**

  Figure 6.16 shows the Usage History page of the system. This page provides a comprehensive record of all materials that have been issued and used across different construction sites. It plays a vital role in tracking material consumption and ensuring transparency in system operations.

  The page displays detailed information such as material name, category, quantity used, unit of measurement, issued user, site, and date of issue. This structured data helps the admin easily monitor how materials are being utilized and identify patterns of usage. It also supports better planning by providing insights into material consumption over time.

  Additionally, the Usage History page ensures accountability by maintaining a clear record of which user accessed or used specific materials. This reduces the chances of misuse and helps in maintaining proper control over resources. The availability of historical data also assists in auditing and reporting purposes.

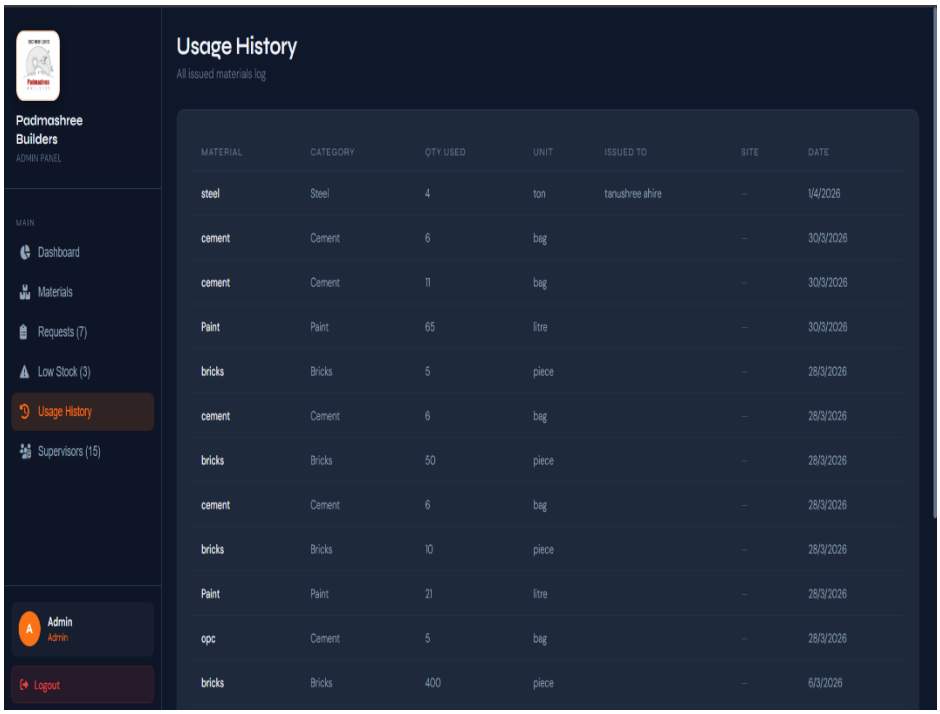

Figure 6.16: Usage History Page Material Usage Tracking

Table 6.15: Usage History Page — Material Usage Tracking validation

Fields	Validation Applied
Material Name	Displays name of the material used
Category	Shows material category
Quantity Used	Displays quantity issued/used
Unit	Shows unit (e.g., ton, bag, piece, litre)
Issued To	Displays name of supervisor/user
Site	Shows site information (if available)

Date	Displays date of material usage
Data Accuracy	Data fetched in real-time from database
Record Tracking	Maintains complete history of material usage

Main Outcomes:

- **Complete Usage Tracking:** Maintains a detailed record of all issued materials
- **Transparency:** Ensures clear visibility of material usage across sites
- **Accountability:** Tracks which user used specific materials
- **Consumption Analysis:** Helps analyze material usage patterns over time
- **Improved Monitoring:** Supports better control over inventory usage

6.17 Supervisors Page

Figure 6.17 shows the Supervisors page of the system. This page allows the admin to manage all supervisor accounts and their assigned sites within the Site Material Management System. It serves as an important module for handling user management and ensuring proper assignment of responsibilities.

The page displays detailed information such as supervisor name, email address, username, assigned site, and account creation date. This organized view helps the admin easily monitor and maintain supervisor records.

In addition to viewing details, the admin can perform various actions such as adding new supervisors, editing existing information, or deleting supervisor accounts when required. These functionalities help in maintaining accurate and up-to-date user data.

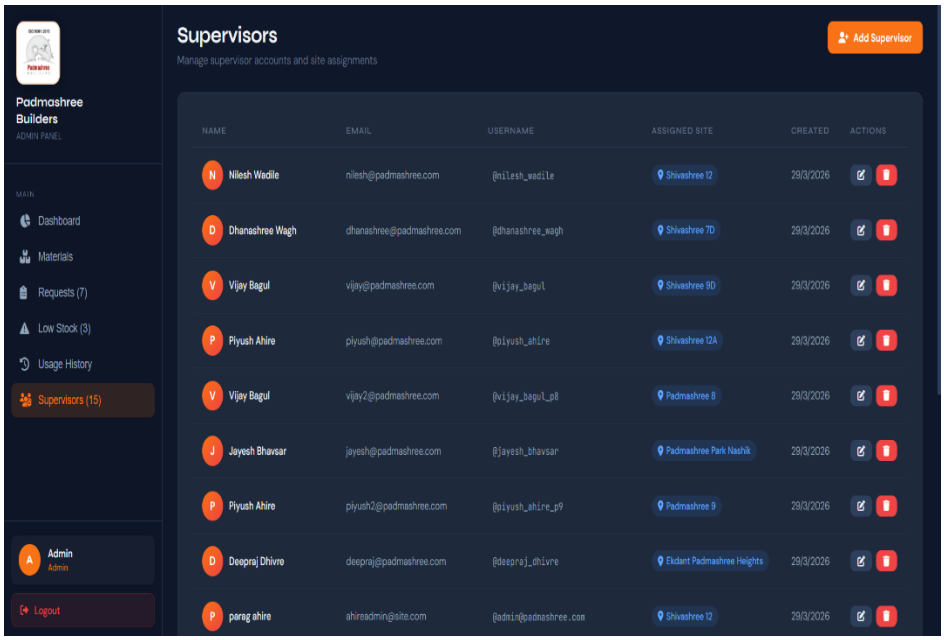

Figure 6.17: Supervisors Page — Supervisor Management and Site Assignment

Table 6.16: Supervisors Page — Supervisor Management and Site Assignment Validation.

Fields	Validation Applied
Supervisor Name	Displays name of the supervisor
Email	Must be in valid email format and unique
Username	Must be unique for each supervisor
Created Date	Displays account creation date
Add Supervisor Button	Allows admin to create new supervisor account
Edit Action	Enables updating supervisor details

Unique Validation	Prevents duplicate email or username
Data Accuracy	Data fetched and updated in real-time from database

Main outcomes:

- **Efficient User Management:** Allows admin to manage all supervisor accounts in one place
- **Proper Site Assignment:** Ensures supervisors are assigned to correct construction sites
- **Organized Records:** Maintains structured and up-to-date supervisor information
- **Easy Account Control:** Supports adding, editing, and deleting supervisor accounts
- **Data Accuracy:** Ensures unique email and username for each supervisor.

6.18 Add New Supervisor Form

Figure 6.18 shows the "Add New Supervisor" form used by the admin to create a new supervisor account in the Site Material Management System. This form plays a key role in managing user access and ensuring that supervisors are properly assigned to their respective construction sites. The admin is required to enter essential details such as full name, email address, username (optional), assigned site, and password. These details help in uniquely identifying each supervisor and linking them to a specific site for managing material-related operations. Assigning the correct site ensures that supervisors can access only the relevant data and perform their tasks efficiently. The form is designed with proper validation mechanisms to ensure accuracy and security

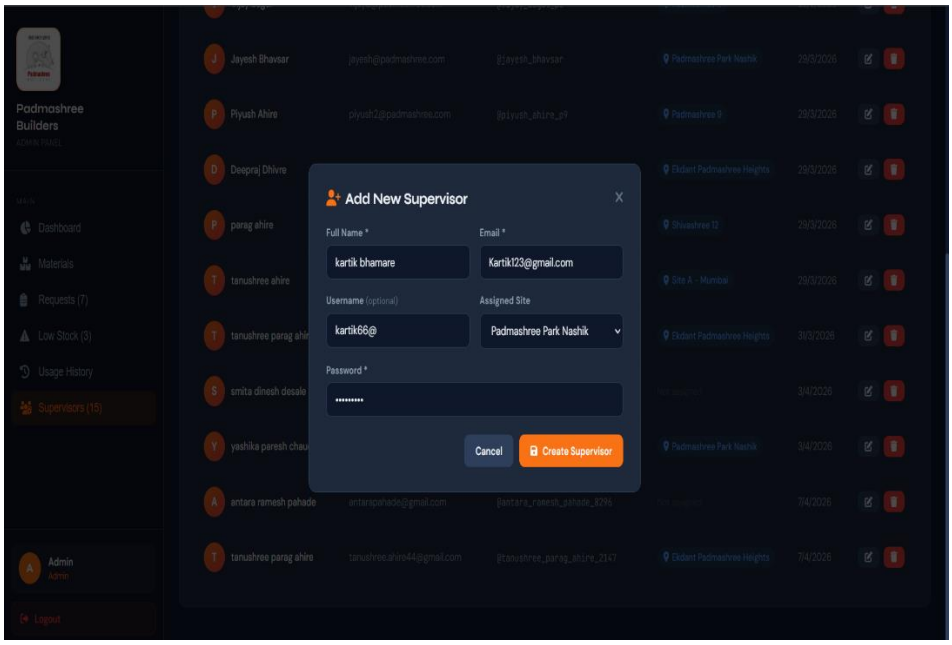

Fig. 6.18: Add New Supervisor Form — Supervisor Account Creation Validations

Table 6.17: Add New Supervisor Form — Supervisor Account Creation Validation

Fields	Validation Applied
Full Name	Must not be empty; accepts alphabetic characters
Email	Must be in valid email format and unique
Username	Optional; if provided, must be unique
Assigned Site	Must be selected from dropdown
Password	Must not be empty; minimum length required

Empty Fields Validation	Required fields cannot be empty
Create Supervisor	Enabled only when all required fields are valid
Cancel Button	Closes the form without saving
Data Submission	Stores supervisor details in database after validation

Main Outcomes

- **Easy Supervisor Creation**: Allows admin to quickly add new supervisor accounts
- **Accurate Data Entry:** Ensures correct input through proper validation
- **Secure Account Setup:** Maintains secure credentials for each supervisor
- **Proper Site Assignment:** Links supervisors to specific construction sites

CHAPTER 7

TESTING AND EVALUATION

7.1 INTRODUCTION

This chapter covers the testing activities performed for the Dhule Padmashree Builders Site Material Management System a web-based application developed to manage construction materials efficiently for organizations and sites.

Testing is a crucial phase of the Software Development Life Cycle (SDLC) that ensures the system works correctly, meets all specified requirements, and is free from defects. The main objective of this testing phase is to:

- Verify that the user authentication and login system works

correctly.

- Ensure that material management operations (add, update, delete) function as per requirements.
- Validate the inventory management and stock tracking system.
- Check the material request and approval system between users and admin.
- Confirm that the report generation module works properly.
- Identify and fix bugs before final deployment.

Testing was performed for both:

- **Positive scenarios** (valid inputs, successful operations)
- **Negative scenarios** (invalid inputs, errors, unauthorized access)

Testing Tools Used:

- **Selenium IDE:** Used for recording and testing UI functionality.
- **Selenium WebDriver**: Used for automated testing of web application features.
- **Postman:** Used for testing backend APIs and server responses.

Modules covered in this chapter:

- User Authentication Module
- Material Management Module
- Inventory Management Module
- Material Request Module (Core Feature)
- Admin Approval Module (Core Feature)
- Report Generation Module
- Product Comparison Module (Core Feature)
- Get a Quote Module (Core Feature)

Out of scope for this chapter:

- Payment gateway testing (not implemented in this version of

the system).

- Real-time material tracking (live GPS/location tracking) is not included in this system.
- Advanced analytics and forecasting features are not covered.
- Mobile application testing (system is web-based only).
- Performance testing under heavy load conditions is not included in this phase.

7.2 Test Plan

The test plan defines the objectives, scope, testing environment, tools, entry and exit criteria, and risk factors associated with the testing cycle of the Site Material Management System. This test plan has been prepared following IEEE standard format to ensure a well-structured and standardized documentation approach.

Table 7.1: Test Plan

Test Plan ID	TP_SMMS_01
Project Name	Site Material Management System
Version	1.0
Date	April 2026

1. **Introduction**

The Site Material Management System is a web-based application developed to manage construction materials efficiently. In traditional methods, material records are maintained manually, which often leads to errors, data loss, and difficulty in tracking inventory.

This system helps in digitalizing the entire process, making it easier to store, update, and manage material data in a structured way.

The system allows users to perform operations such as adding, updating, and deleting materials, checking stock availability, and managing material requests. It improves accuracy, reduces manual effort, and ensures better control over material usage.

The application is built using modern technologies like React.js, Node.js, and MongoDB to provide a user-friendly solution.

1. **Test Items**
   - Login form UI – input fields, submit button, error messages
   - Registration form and user authentication system
   - Material add/update/delete with stock management
   - Inventory dashboard – stock details and tracking
   - Material request module – request submission and status
   - Admin approval/rejection of material requests
   - Report generation module – generate and view reports
   - Admin dashboard – manage users, materials, and requests
2. **Features to be Tested**
   - User Authentication: Register, Login, Logout
   - Role-based access (Admin & User)
   - Material CRUD operations (Create, Read, Update, Delete)
   - Inventory management and stock tracking
   - Material request submission and processing (core feature)
   - Admin approval/rejection system (core feature)
   - Report generation functionality
   - Input validation and error handling
3. **Features Not to be Tested**
   - Online payment system (not implemented)
   - Real-time GPS-based material tracking
   - Cloud storage integration for files/images
   - Heavy load/performance testing (50+ users)
4. **Approach**

- Functional and non-functional testing will be performed
- Both positive and negative test cases will be executed
- Manual testing, Selenium IDE, Selenium WebDriver, and Postman will be used
- UI validation testing for all forms
- Role-based access testing for Admin and User

5. **Pass / Fail Criteria**
   - All important test cases must pass successfully
   - Minimum 95% requirement coverage
   - No critical or blocking bugs at final stage
   - If major bugs are found, testing will pause until fixed
   - Expected output must match actual output
   - All major bugs must be fixed and retested
6. **Suspension Criteria and Resumption Requirements**
   - Testing will stop if critical bugs affect core features
   - Testing will resume after bug fixing and system stabilization
7. **Test Deliverables**
   - Test plan document
   - Test cases document (for all 7 modules)
   - Defect report (per defect found)
   - Test summary report
8. **Test Tasks**
   - Prepare test plan and test cases for all modules
   - Execute test cases in the test environment
   - Identify, log, and track defects
   - Retest fixed defects and update defect status
   - Prepare final test summary report
9. **Test Environment**
   - Operating System: Windows 11
   - Browser: Google Chrome (v90+), Microsoft Edge (v90+)
   - Frontend: React.js + Tailwind CSS (Vite, localhost)
   - Backend: Node.js + Express.js
   - Database: MongoDB (Mongoose ODM)

- Authentication: JWT (JSON Web Tokens)
- Email Service: emailer (Gmail SMTP)
- File Storage: Multer (Local Disk /uploads)
- Server: Localhost / Test Server
- Tools: Selenium IDE, Selenium WebDriver, Postman
- Network: Active internet connection required

10. Responsibilities

- Tester: Execute test cases and report bugs
- Developer: Fix bugs and perform unit testing
- Project Manager: Approve release decision based on test summary

11. Staffing and Training

- Basic knowledge of system modules, web testing, Selenium tools, and Postman is required. Understanding of MongoDB and authentication is beneficial.

12. Scheduling

- Project Deadline: April 2026

13. Risks

- Incomplete test data or missing test accounts for all three user roles
- Gmail SMTP dependency for email notification testing
- Browser compatibility issues on different Chrome/Edge versions
- Changing product requirements during the testing cycle
- Multer file upload errors affecting product image test cases

14. Approvals

- Project Guide
- Team Members

7.2.1 Test Cases

Table 7.2: Admin Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_AL_01	Verify admin login page display	Open Admin Login page	Admin login page should be displayed	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_AL_02	Valid admin login	Enter valid admin email & password	Redirect to admin dashboard	As expected	Pass
TC_AL_03	Invalid password	Enter valid email + wrong password	Error message displayed	As expected	Pass
TC_AL_04	Invalid email	Enter incorrect email	Error message displayed	As expected	Pass
TC_AL_05	Empty fields validation	Submit without entering data	Validation message should appear	As expected	Pass
TC_AL_06	Only email entered	Enter email, leave password	Password required message shown	As expected	Pass
TC_AL_07	Only password entered	Enter password, leave email	Email required message shown	As expected	Pass
TC_AL_08	Security validation	Enter special characters/SQL input	Input should be rejected or handled securely	As expected	Pass

Table 7.3: Home Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_HP_01	Verify home page loads	Open application URL	Home page should display successfully	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_HP_02	Verify navigation links	Click on Materials/About/Contact	Respective pages should open	As expected	Pass
TC_HP_03	Verify login redirection	Click Login button	Redirect to login page	As expected	Pass

Table 7.4: Login Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_LG_01	Valid login	Enter valid email & password	Redirect to dashboard	As expected	Pass
TC_LG_02	Invalid password	Enter valid email + wrong password	Error message displayed	As expected	Pass
TC_LG_03	Invalid email	Enter wrong email	Error message displayed	As expected	Pass
TC_LG_04	Empty fields	Leave all fields blank	Validation message shown	As expected	Pass
TC_LG_05	Only email entered	Enter email, leave password	Password required message	As expected	Pass
TC_LG_06	Only password entered	Enter password only	Email required message	As expected	Pass
TC_LG_07	Role redirection	Login as admin/supervisor	Redirect to correct dashboard	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_LG_08	SQL injection check	Enter special characters	Input rejected	As expected	Pass

Table 7.5: Check Email Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_CE_01	Verify message display	Complete registration	Message to check email shown	As expected	Pass
TC_CE_02	Verify email sent	Check registered email	Credentials received	As expected	Pass

Table 7.6: Site Selection Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_SS_01	Display assigned sites	Login as supervisor	Assigned sites visible	As expected	Pass
TC_SS_02	Select site	Click on site	Redirect to dashboard	As expected	Pass

Table 7.7: Available Materials Page Module

Test Case ID	**Objective**	**Steps (Input)**	**Expected Output**	**Actual Output**	**Status**
TC_AM_01	View materials	Open page	Materials list displayed	As expected	Pass
TC_AM_02	Request material	Click “Request”	Form should open	As expected	Pass
Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_MR_01	View requests	Open page	Request list displayed	As expected	Pass
TC_MR_02	Check status	View request status	Correct status shown	As expected	Pass

Table 7.8: My Request Page Module

Test Case ID	**Objective**	**Steps (Input)**	**Expected Output**	**Actual Output**	**Status**
TC_MR_01	View requests	Open page	Request list displayed	As expected	Pass
TC_MR_02	Check status	View request status	Correct status shown	As expected	Pass

Table 7.9: Request Material Form Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_RM_01	Submit request	Enter details & submit	Request stored successfully	As expected	Pass
TC_RM_02	Validate input	Leave fields empty	Error message displayed	As expected	Pass

Table 7.10: Low stock alert Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_LSA_01	View low stock	Open page	Low stock items shown	As expected	Pass
TC_LSA_02	Alert message	Check alert text	Correct message shown	As expected	Pass
TC_LSA_03	Highlight check	View low items	Highlighted	As expected	Pass
TC_LSA_04	Update stock	Click update	Stock increases	As expected	Pass
TC_LSA_05	Threshold logic	Compare values	Correct detection	As expected	Pass
TC_LSA_06	No low stock	All stock sufficient	No alert shown	As expected	Pass
TC_LSA_07	Data accuracy	Verify numbers	Correct values	As expected	Pass
TC_LSA_08	Navigation	Go to materials page	Works correctly	As expected	Pass

Table 7.11: Add new Material Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_ANM_01	Verify form display	Click “Add New Material”	Form should open successfully	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_ANM_02	Add valid material	Enter all valid details and submit	Material should be added successfully	As expected	Pass
TC_ANM_03	Empty field validation	Submit form without entering data	Validation error messages should be displayed	As expected	Pass
TC_ANM_04	Invalid quantity	Enter negative/invalid quantity	Error message should be shown	As expected	Pass
TC_ANM_05	Invalid price	Enter non-numeric or negative price	Error message should be displayed	As expected	Pass
TC_ANM_06	Threshold validation	Enter invalid threshold value	Validation error should appear	As expected	Pass
TC_ANM_07	Duplicate material entry	Enter existing material name	System should prevent duplication or warn user	As expected	Pass
TC_ANM_08	Cancel operation	Click cancel button	Form should close without saving data	As expected	Pass

Table 7.12: Supervisor Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_SP_01	Verify page display	Login as admin → Open Supervisors page	Supervisors page should be displayed	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_SP_02	Verify supervisors list	View supervisors table	List of all supervisors should be visible	As expected	Pass
TC_SP_03	Verify data accuracy	Check supervisor details	Name, email, username, site should be correct	As expected	Pass
TC_SP_04	Verify add supervisor	Click “Add Supervisor” and submit details	New supervisor should be added	As expected	Pass
TC_SP_05	Verify edit supervisor	Edit supervisor details	Changes should be updated successfully	As expected	Pass
TC_SP_06	Verify delete supervisor	Delete a supervisor account	Supervisor should be removed from list	As expected	Pass
TC_SP_07	Verify unique validation	Add supervisor with same email	System should prevent duplication	As expected	Pass
TC_SP_08	Verify page refresh	Reload page	Updated supervisor list should persist	As expected	Pass

Table 7.13: Add New Supervisor Page Module

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_AS_01	Verify form display	Click “Add New Supervisor”	Form should open successfully	As expected	Pass

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_AS_02	Add valid supervisor	Enter all valid details and submit	Supervisor account should be created	As expected	Pass
TC_AS_03	Empty field validation	Submit form without entering data	Validation error messages should be displayed	As expected	Pass
TC_AS_04	Email format validation	Enter invalid email format	Error message should be shown	As expected	Pass
TC_AS_05	Duplicate email check	Enter existing email	System should prevent duplication	As expected	Pass
TC_AS_06	Password validation	Enter weak/invalid password	Error message should be displayed	As expected	Pass
TC_AS_07	Site assignment	Select site and submit form	Supervisor should be assigned to selected site	As expected	Pass
TC_AS_08	Cancel operation	Click cancel button	Form should close without saving data	As expected	Pass

3 DEFECT REPORTS

The defect report is a crucial document used to record and track defects identified during the testing process of the Site Material Management System. It provides detailed information about each defect, including its description, severity, priority, steps to reproduce, and current status.

This report helps the development and testing teams to understand, prioritize, and resolve issues efficiently. It ensures effective communication among team members and maintains transparency throughout the defect management process

Defect Report 1

Table 7.14: Defect Report 1

Defect ID	DR_SMMS_001
Project Name	Site Material Management System
Product	Web Application
Release Version	v1.0
Module	User Authentication Module — Forgot Password
Detected Build Version	v1.1
Summary	System allows login with empty input fields
Description	While testing the login functionality, it was observed that the system allows users to click the login button without entering email and password. The system does not display validation messages and may proceed with invalid input, which can cause unexpected behaviour
Steps to Reproduce	1. Open login page 2. Leave email and password fields empty 3. Click login button 4. Observe system behaviour
Expected Result	System should display validation messages and prevent login with empty fields
Actual Result	No validation message is shown; login attempt is processed
Severity	Medium
Priority	High (Affects real users, needs immediate fix)

Reported By	Test Engineer
Status	Open

Defect Report 2

Table 7.15: Defect Report 2

Defect ID	DR_SMMS_002
Project Name	Site Material Management System
Product	Web Application
Release Version	v1.0
Module	Material Management Module — Add Material
Detected Build Version	v1.1
Summary	Material is added with negative quantity value
Description	During testing of the add material functionality, it was observed that the system accepts negative values for quantity. This leads to incorrect stock data and affects inventory accuracy.
Steps to Reproduce	1. Login as user 2. Go to Add Material page 3. Enter material details 4. Enter quantity as negative value (e.g., -10) 5. Submit form
Expected Result	System should restrict negative values and show validation error
Actual Result	Material is added with negative quantity
Severity	High

Priority	High
Reported By	Test Engineer
Status	Open

Defect Report 3

Table 7.16: Defect Report 3

Defect ID	DR_SMMS_003
Project Name	Site Material Management System
Product	Web Application
Release Version	v1.0
Module	Material Request Module — Approval System
Detected Build Version	v1.1
Summary	Unauthorized user can access admin approval page
Description	While testing role-based access control, it was observed that a normal user can directly access the admin approval page by entering the URL. This is a security issue as only admin should have access.
Steps to Reproduce	1. Login as normal user 2. Manually enter admin approval page URL 3. Press Enter 4. Observe access
Expected Result	Access should be restricted to admin only

Actual Result	User is able to access admin page
Severity	High (Security vulnerability — risk of brute-force attack and unauthorized access)
Priority	High (Needs immediate fix to protect all user accounts)
Reported By	Test Engineer
Status	Open

7.4 TEST SUMMARY REPORT

This section provides a consolidated summary of all test cases executed across all modules of the Site Material Management System, along with the overall quality assessment of the system.

Table 7.17 : Test Summary Report

Module	Total TCs	Passed	Failed	Pass %
User Authentication Module	10	8	2	80%
Material Management Module	8	7	1	87.5%
Inventory Management Module	7	7	0	100%
Material Request Module	8	7	1	87.5%
Admin Approval Module	5	5	1	83%
Report Generation Module	6	6	0	100%

Admin Dashboard Module	6	6	0	100%
TOTAL	**50**	45	5	**90.%**

Overall Quality Assessment

A total of 50 test cases were executed across 7 modules of the Site Material Management System. Out of these, 45 test cases passed and 5 test cases failed, resulting in an overall pass rate of 90%.

i. DR_SMMS_001: Login validation issue (Medium severity — Open)

ii. DR_SMMS_002: Negative quantity accepted in material entry (High severity — Open)

iii. DR_SMMS_003: Unauthorized access to admin module (High severity — Open)

iv. Additional minor validation issues in request module

v. UI inconsistency in approval module

The Core modules such as Inventory Management, Report Generation, and Admin Dashboard achieved a 100% pass rate, indicating stable performance in key functionalities. workflows. The system is functionally stable for major operations. However, the identified defects should be resolved before final deployment to ensure security, accuracy, and reliability

Testing Phase:

The testing phase is an important part of the system development process. After the development of the Site Material Management System, the system was thoroughly tested to ensure that all features and functionalities work correctly and efficiently.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

The Site Material Management System is developed to simplify and improve the process of managing construction materials in an organization or site. This system provides a centralized platform where users can manage materials, track inventory, and send material requests efficiently. It reduces manual work, improves accuracy, and helps in better decision-making. The system includes important features such as:

These features help in maintaining proper records and avoiding material shortages or wastage. Overall, the system successfully meets its objectives and provides an efficient solution for managing site materials digitally.

FUTURE SCOPE:

Mobile Application Development

A mobile app can be developed for easy access using smartphones.

Online Payment Integration

Payment features can be added for purchasing materials online.

Real-Time Order Tracking

Live tracking of material delivery can be implemented.

Advanced Search and Filtering

Graphs and reports can be added for better decision-making.

Cloud Integration

Data can be stored on cloud platforms for better scalability and security

CHAPTER 9

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